

36 Cancer of the Liver



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INTRODUCTION

Primary liver cancers represent the fifth most common malignancy worldwide and the second most common cause of death from cancer. This is mainly due to the association between hepatocellular carcinoma (HCC) and chronic hepatitis B virus (HBV) and hepatitis C virus (HCV) infections.¹ Untreated HCC has a dismal prognosis, with a <10% 5-year survival rate.² The combination of cancer and chronic liver disease adds significant complexity to treatment. Remarkable progress in understanding the natural history, pathogenesis, and tumor biology of HCC has resulted in effective treatment options. For localized HCC, surgical resection, orthotopic liver transplantation (OLT), and ablative therapies are curative approaches. Formidable image-guided local and regional therapies, radiation therapies, and systemic therapies now round out a full arsenal of treatments even for disseminated disease. Recently, there have been marked advances in therapies for hepatitis and in care of the associated liver parenchymal disease that may reduce the incidence of HCV-related HCC. Furthermore, breakthroughs in systemic immunotherapy have also been reported.

EPIDEMIOLOGY

The annual number of worldwide liver cancer cases (748,300) closely resembles the number of deaths (695,900). Long-term survival rates are 3% to 5% in most cancer registries.¹ Globally, HCC is 2.3 times more common in males, likely due to androgen receptors on some of these tumors.³ There has been a significant increase in the incidence of HCC in the US during the past 25 years.⁴ This parallels the increase in HCV infection, the increase in immigrants from HBV-endemic countries, and an increase in nonalcoholic fatty liver disease (NAFLD). However, the introduction of widespread HBV vaccination, such as in Taiwan in 1984, has led to a reduction of liver cancer from 0.54 per 100,000 to 0.2 per 100,000 over a 16-year period.⁵

ETIOLOGIC FACTORS

Viral Hepatitis and Hepatocellular Carcinoma

Both case-control studies and cohort studies have shown a strong association between chronic HBV and HCC. Beasley⁶ followed Taiwanese male postal carriers who were hepatitis B surface antigen (HBsAg) positive and found a 98-fold greater risk of HCC compared to HBsAg-negative individuals. For asymptomatic HBsAg-positive blood donors at American Red Cross centers, a relative risk of 12.7 was noted for liver cancer compared with HBsAg-negative individuals. In addition to hepatitis virus levels, factors predictive of HCC include male gender, advanced age, specific promoter mutations, and the presence of cirrhosis.⁷

The exact mechanism by which HBV infection causes HCC is not known. Some have postulated that the effect of HBV on hepatic carcinogenesis is indirect, through the process of inflammation, regeneration, and fibrosis associated with chronic hepatitis and cirrhosis. Consistent with this hypothesis, 70% of HBV-related HCC cases occur in association with cirrhosis. There is also evidence for a direct viral effect. Integration of HBV DNA has been observed within the retinoic acid receptor alpha gene and within the human cyclin A gene, both of which play crucial roles in cellular growth. The hepatitis Bx (HBx) gene product is implicated in causing HCC. It is a transcriptional activator of various cellular genes associated with growth control. HBx interacts with *p53*, interfering with its tumor suppressor activity.

HCV is an RNA virus without a DNA intermediate form and therefore cannot integrate into hepatocyte DNA. In contrast to HBV, HCV is more likely to lead to chronic infection (10%: HBV vs 60% to 80%: HCV) and cirrhosis (20-fold increase).⁸ The typical interval between HCV-associated transfusion and subsequent HCC is only about 30 years (compared to 40 to 50 years for HBV). HCV-related HCC patients tend to have more frequent and advanced cirrhosis. In HBV-related HCC, only half the patients have cirrhosis. The risk of developing HCC with HCV-related cirrhosis is 5%/year versus 0.5%/year for HBV-related cirrhosis.

There have been extensive efforts to establish the molecular pathways involved in the pathogenesis of HCC.⁹ Abnormalities commonly found in HCC cells include¹ cell-cycle dysregulation associated with somatic mutations or loss of heterozygosity in *TP53*, silencing of *CDKN2A* or *RBI*, or *CCND1* overexpression²; angiogenesis accompanied by overexpression or amplification of *VEGF*, *PDGF*, and *ANGPT2*³; evasion of apoptosis as a result of activation of survival signals such as nuclear factor kappa B (NF- κ B); and⁴ reactivation of *TERT*.

There is emerging evidence of the importance of microRNAs (miRs) and epigenetic alterations such as hypermethylation in the pathogenesis of HCC. miR-155 levels are significantly increased in patients infected with HCV, and overexpression of miR-155 is associated with nuclear accumulation of beta-catenin by increased Wnt signaling, thereby implicating this pathway in HCV-related HCC.¹⁰

Alcohol-Induced Hepatocarcinogenesis

Chronic alcohol consumption leads to oxidative stress in the liver, inflammation, cirrhosis, and development of HCC. Ethanol is metabolized by alcohol dehydrogenases and cytochrome P-450, producing acetaldehyde and reactive oxygen species (ROS). Acetaldehyde binds directly to proteins and DNA and damages mitochondria, initiating apoptosis. P-450 metabolism leads to ROS generation, which leads to lipid consumption and peroxidation, protein oxidation, and formation of DNA adducts.¹¹ Alcohol leads to monocyte activation and inflammatory cytokine production. Oxidative stress promotes the development of fibrosis and cirrhosis, creating a permissive HCC microenvironment. It may also lead to decreased Signal Transducers and Activators of Transcription 1

(STAT1)–directed activation of interferon-gamma (IFN γ) signaling with consequent hepatocyte damage.¹²

Nonalcoholic Fatty Liver Disease (NAFLD)

HCC is linked to NAFLD, which is present in 30% of the general adult population, 90% of morbidly obese adults (body mass index [BMI] ≥ 40 kg/m²), and close to 74% of those with diabetes.¹³ The risk of HCC due to NAFLD appears to be less than that of chronic hepatitis C. A recent US study reported a 2.6% yearly cumulative incidence of HCC in NAFLD and 4% in HCV cirrhosis.¹⁴ Similarly, a prospective 5-year study from Japan reported an 11.3% HCC rate among NAFLD cirrhosis patients compared to 30.5% among HCV-related cirrhosis patients.¹⁵

A recent study from Germany identified nonalcoholic steatohepatitis (NASH) as the most common etiology of HCC (24%), surpassing chronic hepatitis C (23.3%), chronic hepatitis B (19.3%), and alcoholic liver disease (12.7%).¹⁶ Linoleic acid is a fatty acid that accumulates in NAFLD, causes more oxidative damage than other free fatty acids, mediates the selective loss of CD4⁺ T lymphocytes, and leads to accelerated hepatocarcinogenesis. In an HCC mouse model, in vivo blockade of ROS reversed NAFLD-induced hepatic CD4⁺ T cell loss and delayed NAFLD-induced HCC.

Diabetes and obesity are known independent risk factors for HCC, and this association holds in NAFLD and associated NASH. HCC can develop in patients with metabolic syndrome and NAFLD in the absence of NASH and fibrosis.¹⁷

Other Etiologic Considerations

Other underlying conditions that are associated with the development of HCC are listed in Table 36.1.

Chemical Carcinogens

Probably the best-studied and most potent ubiquitous natural chemical carcinogen is a product of the *Aspergillus* fungus, called aflatoxin B₁.¹⁸ *Aspergillus flavus* mold and aflatoxins can be found mainly in hot, humid parts of the world, where grains such as rice are stored in unrefrigerated conditions. In the months following the monsoons in Southeast Asia, most village-based grains are covered by a white layer of aflatoxin. Data on aflatoxin contamination of foodstuffs correlate well with HCC incidence rates in Africa and China.

There is considerable literature on the hepatocarcinogenicity of anabolic steroids.¹⁹ Although estrogens are capable of causing HCC in rodents, an epidemiologic association in humans has not been demonstrated. Environmental pollutants, particularly pesticides and insecticides, are known rodent hepatic carcinogens. In a recent case-control study, cumulative lifetime tobacco use of more than 11,000 packs and Asian ethnicity were independent predictors of HCC development among chronic liver disease patients.²⁰

Adenomas

Adenomas are benign liver tumors with malignant transformation potential and an incidence of 3 to 4 per 100,000. They are more common in women and associated with oral contraceptives, obesity, and metabolic syndrome. Twenty percent of adenomas are multifocal. Adenomas are classified into five molecular subtypes: hepatocyte nuclear factor-1 alpha (HNF1A) mutated, beta-catenin 1 (CTNNB1), interleukin-6 signal transducer gene mutated, sonic hedgehog activated, and unclassified.²¹ Of these, the beta-catenin 1 subtype has the greatest likelihood of malignant transformation. Other risk factors for malignant transformation include male sex,

TABLE 36.1

Conditions Associated with Human Hepatocellular Carcinoma

Condition	Risk
<i>Cirrhosis</i>	
Hepatitis B virus	High
Hepatitis C virus	High
Alcohol	High
Autoimmune chronic active hepatitis	High
Cryptogenic cirrhosis	High
Cirrhosis due to nonalcoholic fatty liver disease	High
Primary biliary cirrhosis	Low
Hereditary hemochromatosis	High
α_1 -Antitrypsin deficiency	High
Wilson disease	Low
<i>Metabolic Diseases (without Cirrhosis)</i>	
Hereditary tyrosinemia	High
α_1 -Antitrypsin deficiency	Moderate
Ataxia telangiectasia	Moderate
Types 1 and 3 glycogen storage disease	Moderate
Galactosemia	Moderate
Citrullinemia	Moderate
Hereditary hemorrhagic telangiectasia	Moderate
Porphyria cutanea tarda	Moderate
Orotic aciduria	Moderate
Alagille syndrome (congenital cholestatic syndrome)	Moderate
<i>Environmental</i>	
Thorotrast	Moderate
Androgenic steroids	Moderate
Cigarette smoking	Low to moderate
Aflatoxin	Moderate
<i>Precancerous conditions</i>	
Adenoma	Moderate

glycogen storage disease, anabolic steroids, androgen therapy, and tumors >5 cm in diameter.

DIAGNOSIS

HCC is diagnosed by using radiologic studies and pathologic diagnosis with a biopsy. Core biopsies are most preferred because of the tissue architecture given by this technique. A core biopsy of the portal vein may be performed for patients suspected of having portal vein involvement. Morphologic features, such as stromal invasion, help distinguish high-grade dysplastic nodules from HCC.²²

The American Association for the Study of Liver Diseases (AASLD) and the European Association for the Study of the Liver (EASL) have outlined noninvasive criteria for HCC diagnosis. The current EASL guidelines recommend lesions >1 cm with characteristic radiologic features of arterial hyperenhancement on two different imaging modalities with washout, and no biopsy is needed.²³ The AASLD guidelines now defer to the LI-RAD

recommendations.^{23,24} Bialecki et al.²⁵ found a sensitivity and specificity of 89.1% and 100%, respectively, for liver biopsy compared to 64.9% and 62.8%, respectively, for the noninvasive EASL criteria. There is fear of biopsy-related hemorrhage and tumor seeding associated with biopsies. However, data show that hemorrhage occurs at only a 0.4% rate, and tumor seeding occurs at a low rate of 1.6%. When seeding occurs, it can be treated by local resection and is seldom a cause of morbidity and mortality.

STAGING

Multiple clinical staging systems for hepatic tumors are described. The most widely used is the American Joint Committee on Cancer/tumor-node-metastasis (AJCC/TNM) (Table 36.2A).²⁶ Adverse prognostic features include large size, multiple tumors, vascular invasion, and lymph node spread. Macroscopic or microscopic vascular invasions, in particular, have profound effects on prognosis. Stage III disease contains a mixture of lymph node-positive and negative tumors. Stage III patients with positive lymph nodes or stage IV disease have a poor prognosis, and few patients survive one year.

The prognosis in HCC patients is influenced by the presence and severity of the underlying liver disease. The Child-Turcotte-

TABLE 36.2A

Staging for Liver Function and Cancer: American Joint Commission on Cancer Staging for Hepatocellular Cancer

T	TX	Primary tumor cannot be assessed		
	T0	No evidence of primary tumor		
	T1	Solitary tumor <2 cm, or >2 cm without vascular invasion		
	T1a	Solitary tumor <2 cm		
	T1b	Solitary tumor >2 cm without vascular invasion		
	T2	Solitary tumor >2 cm with vascular invasion; or multiple tumors, none >5 cm		
	T3	Multiple tumors, at least one of which is >5 cm		
	T4	Single tumor or tumors of any size involving a major branch of the portal vein or hepatic vein, or tumor(s) with direct invasion of adjacent organs other than the gallbladder or with perforation of visceral peritoneum		
N	NX	Regional lymph nodes cannot be assessed		
	N0	No regional lymph node metastasis		
	N1	Regional lymph node metastasis		
M	MX	Distant metastasis cannot be assessed		
	M0	No distant metastasis		
	M1	Distant metastasis		
Stage Grouping				
	IA	T1a	N0	M0
	IB	T1b	N0	M0
	II	T2	N0	M0
	IIIA	T3	N0	M0
	IIIB	T4	N0	M0
	IVA	Any T	N1	M0
	IVB	Any T	Any N	M1

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TABLE 36.2B

Child-Pugh Grading of Cirrhosis^a

Measurement	1 Point	2 Points	3 Points
Bilirubin (mg/dL)	1–1.9	2–2.9	>2.9
Prolongation of PT	1–3	4–6	>6
Albumin (g/dL)	>3.5	2.8–3.4	<2.8
Ascites	None	Mild	Moderate/severe
Encephalopathy	None	Grade 1 or 2	Grade 3 or 4

^aGrade A = 5–6 points; grade B = 7–9 points; grade C = 10–15 points. PT, prothrombin time.

Pugh (CTP) scoring system is most commonly used for assessing cirrhosis (Table 36.2B). It encompasses five parameters—bilirubin, albumin, prothrombin time, clinical ascites, and clinical encephalopathy—each of which is scored from one to three depending on severity. The key limitation of this system is its lack of any parameters pertaining to the cancer itself. Despite that, it remains incorporated into many HCC clinical trials as a tool to measure the extent of liver disease in the study populations. However, this main limitation has been overcome by other scoring systems. Among those, the first to be established is the Okuda staging system. The Cancer of the Liver Italian Program (CLIP) score was defined and studied prospectively in mainly HCV-related HCC patients. The CLIP score consists of the CTP score parameters combined with a subjective assessment of the tumor in the liver, the presence or absence of portal vein thrombosis (PVT), and the alpha-fetoprotein (AFP) level. The addition of vascular endothelial growth factor (VEGF) levels to the CLIP parameters (V-CLIP) has provided a significantly more precise prognosis but has yet to be prospectively validated.²⁷ The Chinese University Prognostic Index (CUPI) scoring system was developed in patients with mainly HBV-related HCC.²⁸ The CUPI parameters are bilirubin, ascites, AFP, alkaline phosphatase, the tumor extent (AJCC/TNM 5th edition), and clinical symptoms at presentation. A French system called the Groupe d'Etude et de Traitement du Carcinoma Hepatocellulaire (GETCH) staging system consists of bilirubin, Karnofsky performance score, AFP, alkaline phosphatase, and PVT.²⁹ Another scoring system mainly used in Japan is the Japan Integrated Staging (JIS) score.³⁰

Another commonly used scoring system is the Barcelona Clinic Liver Cancer (BCLC) classification system.³¹ The BCLC couples prognosis with treatment assignment and has been validated prospectively. However, it is less valuable in the setting of more advanced disease, defined as BCLC category C. In retrospective analyses of advanced-stage HCC patients seen by medical oncologists, the CLIP scoring system was the most informative regarding the outcome of this specific patient population.

More recently, a simple analysis of Albumin-and Bilirubin (ALBI score) has shown similar prognostication value as much more complex scoring systems. This simplified ALBI scoring system has also been validated with large Eastern and Western datasets of patients subjected to surgery, chemoembolization, or systemic biologic therapies.³²

MOLECULAR AND BIOMARKER

Mutational Landscape of Genetic Alterations in HCC

Many genetic changes have been documented in HCC. Most of these are related to cell growth and angiogenesis, both processes important in transformation, progression, and metastases. Many of the genetic changes well documented in HCC are listed in

TABLE 36.3

Major Functional Signaling Pathways and Molecules in Hepatocellular Carcinoma

Functional Process (Frequency)	Signaling Pathway	Associated Molecules (Frequency %)	Mode of Action in Tumors	Phenotypic Features
Cell cycle (4%–35%)	p53, RB–E2F	TP53, CKN2A/B, CCND/E1, CDKs, ATM, RB1	Inhibition	Loss during tumor progression, aggressive phenotype, DNA damage repair mechanisms
	Wnt–β-catenin	CTNNB1, AXIN1+2, APC	Activation	Early and late stage, genomic stability, activation in tumor-initiating cells
	TGF-β	SMADs	Early: inhibition Late: activation	Poor prognosis, metastasis, tumor-initiating cells
Development and differentiation (2%–55%)	Hippo–YAP	Growth control, via YAP and MST1/2	Activation	Stem cell features, tumor initiation, chemoresistance
	NF2	Growth control, via YAP, EGFR	Inhibition	Stem cell features, tumor initiation
	SALL4	NURD complex, PTEN	Activation	Development and progression, poor prognosis, tumor-initiating cells
	IGF	IGF1R, IRF2, IRS, SHP, PI3K	Activation	Preneoplastic lesions, early stage
	HGF–MET	SH2, MAPK, MEK, ERK	Activation	Metastatic potential, invasion
Oncogenic signaling (0%–35%)	EGFR	AKT, STATs, RAS/RAF	Activation	Aggressive phenotype, reprogramming
	PI3K–mTOR	PIK3CA, AKT, RPS6KA3, PTEN, TSC1, RAPTOR, RICTOR	Activation	Poor prognosis, poorly differentiated tumors, and earlier recurrence
	VEGF	VEGFR, HIF1 α	Activation	Aggressive phenotype, poor prognosis, metastasis
Angiogenesis (0%–15%)	FGF	FGF19, FGFRs, SHP2	Activation	Development or progression
	PDGF	ROS, PI3K, STAT3, MMPs	Activation	Liver cirrhosis, development
Immune response (2%–9%)	NF-κB	IKB, IKK, NEMO, p65, IL-20	Activation	Chronic inflammation, tumor progression
	TWEAK/Fn14	NOTCH, WNT	Activation	Tumor-initiating cells, cell fate decision
	IL-6 signaling	STAT3, LIN28, IL-6R, IL-6, JAK1	Activation	Progenitor derived, response to adjuvant interferon therapy
Posttranscriptional modifications (not reported)	RNA editing	AZIN1, ADARs, ODC	Alteration	Cirrhosis, development and recurrence, poor prognosis
Genome maintenance (10%–28%)	Chromatin remodeling	ARID1 and ARID2, MLL, BAP1, EZH2	Activation	Frequently mutated (SWI/SNF), poor prognosis
	Telomere stability	TERT	Activation	Earliest genetic changes, associated with activation of WNT– β -catenin
Stress response/mitochondria (0%–22%)	Oxidative response	NRF2, KEAP1, CUL3	Activation	Oxidative phosphorylation, tumor progression, late stage

With permission from Marquardt JU, Andersen JB, Thorgeirsson SS. Functional and genetic deconstruction of the cellular origin in liver cancer. *Nat Rev Cancer* 2015;15(11):653–667.³³

Table 36.3.³³ Most important of these are key oncogenic molecules such as transforming growth factor β 1 (TGF β 1), MYC, phosphatidylinositol 3-kinase (PI3K)/AKT, and growth factors VEGF and FGF.

The most integrative view on hepatocarcinogenesis was provided by a recent study of The Cancer Genome Atlas (TCGA) Research Network.³⁴ The TCGA investigators performed multi-layer molecular analyses of whole-exome sequencing and DNA copy number analyses, DNA methylation, RNA, microRNA, and proteomic expression on 363 HCC cases. Whereas the mutational landscape of alterations was comparable to previous studies, significant alterations by hypermethylation in genes likely to result in HCC metabolic reprogramming (ALB, APOB, and CPS1) were observed. Integrative molecular subtyping of five data platforms identified three subclusters of tumors associated with distinct prognostic traits. Furthermore, the authors also revealed potential actionable targets in WNT signaling, MDM4, mesenchymal-ep-

ithelial transition (MET), vascular endothelial growth factor A (VEGFA), myeloid cell leukemia 1 (MCL1), isocitrate dehydrogenase 1 (IDH1), telomerase reverse transcriptase (TERT), immune checkpoint proteins, cytotoxic T-lymphocyte antigen 4 (CTLA-4), programmed cell death protein 1 (PD-1), and programmed cell death protein ligand 1 (PD-L1). These data have provided an inventory of potential targets for clinical development. The results of these studies also highlight the rationale for therapeutic interventions based on the tumor–microenvironment interaction (e.g., by inhibition of checkpoint molecules such as PD-1/PD-L1 and CTLA-4).³⁵

The Microenvironment of Liver Cancer

HCC usually develops in the background of chronic liver disease and, in more than 80% of the cases, in a preexisting liver with cirrhosis.

Recent research has defined an inflammation-fibrosis-cancer axis that contributes to tumor initiation. A 186-gene signature generated from the surrounding nontumoral liver tissue was highly correlated with the outcome of patients in a cohort of more than 300 HCCs.³⁶ Consistently, this poor prognosis signature contained gene sets associated with inflammation such as interferon signaling, NF- κ B, and tumor necrosis factor α . Further, gene set enrichment analysis showed that the downstream targets of IL-6 were strongly associated with the poor prognosis signature, again confirming the importance of this signaling in hepatocarcinogenesis.

Molecular Classification and Prognostic Prediction of Hepatocellular Carcinoma

Many studies have been performed to use molecular profiling in prognostication in HCC. No molecular profile or classification system has made its way to routine clinical use. An interesting attempt to classify HCC was recently introduced by combining phenotypic and molecular information in a large series of 343 resected HCC samples.³⁷ Results demonstrate that different histologic subtypes are associated with clinical and molecular features. Alterations in CTNNB1 and TP53 are mutually exclusive and are associated with distinct morphologic features. Tumors harboring CTNNB1 mutations tend to be larger in size, well differentiated, and without inflammatory features, whereas TP53-mutated tumors are poorly differentiated, pleomorphic, and show vascular invasion. Furthermore, the scirrhous HCC subtype showed alterations in TSC1/TSC2 and more stem-like features, whereas the steatotic subtype showed frequent IL-6/JAK/STAT activation. Interestingly, a novel macrotrabecular-massive subtype showed a poor clinical outcome, high α -fetoprotein (AFP) levels, and FGF19 amplifications. The authors concluded that HCC phenotypes are tightly associated with molecular alterations that may help to translate the current understanding of HCC biology. Histologic and morphologic subtypes predict outcome and correlate to specific genomic profiles. However, no genetic profile or mutation has been found to be a prognostic predictor independent of stage, morphology, and histology. Although many alterations in genomics, epigenetics, and circulating genetic markers have been studied, none have been found to be superior to clinicopathologic features of the tumors and circulating AFP.

Alpha-Fetoprotein

AFP is a protein encoded by the AFP gene on the q arm of chromosome 4. It is produced by the yolk sac and the fetal liver during fetal development and is thought to be the fetal analog of albumin. Serum AFP has been extensively studied as HCC biomarker, since levels are found to be elevated in the circulation of 70% to 90% of patients with HCC. A threshold of 200 ng/ml has been considered diagnostic for HCC if imaging criteria are also met.³⁸ It is also the best circulating indicator of effective treatment and prognosis. AFP level has been associated with recurrence risk in patients receiving liver resection^{39,40} or tumor ablation.^{41,42} AFP values are useful as a selection tool for liver transplantation.⁴³ AFP correlated with tumor progression following transarterial chemoembolization (TACE).⁴⁴ AFP levels are also prognostic after systemic therapies.⁴⁵

Mutational Burden and Microsatellite Instability

An important predictor of efficacy of immune checkpoint inhibitors (ICI) in many cancers including melanoma, lung cancer, and colon cancer are tumor mutational burden (TMB), checkpoint expression, or microsatellite instability.⁴⁶ Consequently, analysis of TMB by whole-exome sequencing, of PD1, PD-L1 or CTLA-4 expression by immunohistochemistry or nanostring analysis are

TABLE 36.4

Treatment Options for Hepatocellular Carcinoma

- Surgery
 - Partial hepatectomy
 - Liver transplantation
- Local ablative therapies
 - Cryosurgery
 - Microwave ablation
 - Ethanol injection
 - Acetic acid injection
 - Radiofrequency ablation
 - Proton or carbon-ion therapy
 - Conformal radiation therapy
 - Stereotactic radiation therapy
- Regional therapies: hepatic artery transcatheter treatments
 - Transarterial chemotherapy
 - Transarterial embolization
 - Transarterial chemoembolization
 - Transarterial yttrium-90 microspheres
 - Transarterial ¹³¹I-lipiodol
 - Palliative low-dose radiation therapy
- Systemic therapies
 - Chemotherapy
 - Targeted therapy (sorafenib)^a
 - Immunotherapy
 - Hormonal therapy
 - Oncolytic virotherapy
 - Supportive care

^aSorafenib is the only systemic therapy with level 1 evidence, proving a survival benefit.

used in other tumor types to determine treatments. In HCC, TMB has not been predictive of response to either nivolumab or pembrolizumab. Therefore, the value of TMB as a predictive marker for efficacy of ICI therapy in HCC is unclear.⁴⁶

TREATMENT OF HEPATOCELLULAR CARCINOMA

Many treatment options for HCC are available (Table 36.4). Resection and liver transplantation (LT) represent the potentially curative options with the longest track record. For small tumors, ablation and radiotherapy (RT) are quite effective and may also be curative.

Surgical Resection for Hepatocellular Carcinoma

Patient Selection

Liver resection is the preferred treatment for noncirrhotic HCC patients. These patients generally have a normal liver function, no portal hypertension, and can tolerate major liver resections with acceptable morbidity and low mortality. The selection of these patients for resection is as for other malignant lesions. Resection should be considered where a complete resection of the tumor is possible while preserving >30% functional liver. If the potential remnant liver volume may be <30%, portal vein embolization (PVE) with or without total liver venous deprivation⁴⁷ and radiation lobectomy⁴⁸ are accepted preoperative methods for increasing the potential remnant liver volume and thereby increasing the number of patients eligible for hepatic resection.

For cirrhotic patients, the primary determinant of outcome and therapy selection is the degree of hepatic dysfunction and portal

TABLE 36.5

Results of Surgical Resection for Hepatocellular Carcinoma

First Author	Year Published	Number of Patients	Cirrhosis	Minor Resections ^a	Mortality	5-Year Overall Survival Rate
Nathan ⁵⁶	2009	788	—	—	—	39%
Yang ⁵⁷	2009	481	77%	—	1.7%	20%–48% ^b
Wang ⁵⁸	2010	438	—	—	7.5%	43%
Peng ⁵⁹	2012	603	88%	25%	—	11.1%
Zhong ⁶⁰	2014	1259	—	—	3.1%	39%
Utsonomiya ⁶¹	2014	13,318 ^c	42%	—	—	55%
Leung ⁶²	2015	176	—	—	—	58% ^d
Qi ⁶³	2015	6551 ^e	—	—	—	54%
Kim ⁶⁴	2016	277	—	—	—	51.8%
Ng ¹³⁵	2017	218	—	80.4%	0.9%	66.5%
Tada ⁶⁶	2017	489	—	—	—	54.2%
Cucchetti ⁷¹	2020	2,523	—	—	—	61.6%

Note: Selected series with more than 200 patients over the past two decades. Areas marked by a dash are not defined.

^aMinor resections are defined as $\leq$ segmentectomy.

^bRange of 5-year overall survival among different subgroups.

^cLarge prospective cohort study.

^d4-year overall survival.

^eMeta-analysis of 32 studies.

hypertension. Traditionally, only compensated cirrhotics (CTP class A) are candidates for hepatic resection, whereas patients with significant hepatic dysfunction (CTP class B or C) are generally not selected for resection because of poor outcomes. Portal hypertension can be indirectly assessed clinically by the presence of splenomegaly, esophagogastric varices, and thrombocytopenia (platelet count $<100,000/\text{mm}^3$) or directly determined by hepatic venous wedge pressures (≥ 10 mmHg). With recent advances in perioperative care, there is growing evidence that liver resection for HCC in well-selected patients with mild portal hypertension is safe and achieves a comparable outcome as patients without portal hypertension.⁴⁹

There is a consensus that the critical remnant liver volume in cirrhotic patients is $>40\%$, and PVE with or without total liver venous deprivation or radiation lobectomy can be considered if the future remnant liver volume is expected to be below this threshold.⁵⁰ A potential advantage of radiation lobectomy is that it results in tumor control in addition to inducing contralateral hepatic hypertrophy. This is the reason that other investigators have employed sequential or even simultaneous arterially directed therapies (ADT) and PVE before definitive surgical resection. The combination of ADT and PVE results in more efficient hypertrophy of the future remnant liver than PVE alone.^{51,52}

In other parts of the world, dynamic liver function tests are also employed to assess suitability for liver resection. These include the indocyanine green (ICG) test and technetium-99m diethylenetriaminepentaacetic acid-galactosyl human serum albumin (technetium-99m galactosyl human serum albumin [99mTc-GSA] scintigraphy). An ICG retention rate of $<14\%$ is a safe indicator for major hepatectomy in cirrhotic patients, whereas retention rates $>20\%$ are considered a contraindication for major hepatectomy.⁵³

Patient medical comorbidities should be considered for any major surgery. Some studies have reported age and gender as independent risk factors for poor outcomes after HCC resection. Other studies indicate that advanced age is more a surrogate for medical fitness and that, with careful selection, elderly patients benefit as much from resection as younger patients.⁵⁴ Comorbidities, as represented by the American Society of Anesthesia (ASA) grade, have been shown to correlate with survival.⁵⁵

Outcomes of Resection

With improving patient selection and perioperative care, the outcome of hepatic resection for HCC has continuously improved during the past two decades. Many large series show that resection is associated with a perioperative mortality rate of $<5\%$ and an overall survival (OS) rate of $>50\%$ (Table 36.5).^{56–66} Many major centers recently reported $<2\%$,^{57,65,67,68} even in cirrhotic patients.

Tumor factors most predictive for outcome are TNM staging at presentation, macro- and micro-vascular invasion, and the number of tumors. Large HCC has a propensity for vascular invasion and intraluminal tumor growth. This is associated with intrahepatic satellite metastases via the portal venous system and small satellite tumors. Intraluminal spread through the hepatic veins leads to pulmonary metastases.

One surgical factor prognostic for outcome is the surgical margin. There is no clear margin size that has been universally agreed upon, but there is consensus on the importance of an R0 resection. Most surgeons prefer at least a 1-cm margin. In one 225-patient study, a 1-cm margin was associated with a 77% 3-year survival versus 21% for those with <1 -cm margin.⁶⁹ In a randomized controlled trial (RCT), a 2-cm margin was associated with decreased recurrence and improved survival.⁷⁰

Studies have also demonstrated improved outcomes for anatomic versus nonanatomic resections for HCC. In a series of 210 patients, 5-year survival rates were 66% for anatomic versus 35% for nonanatomic resections.⁶⁹ Choice of margin and the anatomic approach for cancer clearance must be weighed against better perioperative outcome for limited parenchymal resection in cirrhotic patients.

A recent multicenter study from three continents shows the effectiveness of liver resection and highlights the shortcomings. Data from 2,523 patients undergoing resection for HCC from major institutions in Asia and Europe were analyzed. The 5-year survival was $>60\%$, and the 10-year survival was 39%.⁷¹ In multivariable analysis, the patient's age and the risk of early HCC recurrence (within 2 years) were independent determinants of the chance of cure ($P < 0.001$).

However, many of the patients were not disease free. These data were used to fit statistical cure models, to compare disease-free

survival (DFS) after surgery to the survival expected for patients with chronic hepatitis and/or cirrhosis and the general population (matched by sex, age, race/ethnicity, and year of diagnosis). The cure model suggested that in HCC patients, resection can achieve the same life expectancy as those with chronic liver disease in 26.3% of cases and as the general population in 17.1% of cases.⁷¹

Allogeneic Liver Transplantation for Hepatocellular Carcinoma

Patient Selection

Theoretically, liver transplantation (LT) is the ideal therapy for HCC in cirrhotic patients because it treats cancer and the underlying parenchymal disease. However, early experience with transplants produced dismal results. Bismuth et al.⁷² was one of the first groups to consider that, in advanced disease, the likelihood of systemic disease was so high that recurrence rates, and therefore long-term outcomes, were unacceptably poor. They demonstrated that patients with limited disease (uninodular or binodular <3 cm tumors) had much better outcomes with a transplant than resection (83% 5-year survival versus 18%).⁷²

The landmark works of Mazzaferro et al.⁷³ have defined the most commonly used criteria for selecting HCC patients for transplantation. They defined the Milan criteria for transplantation as a single tumor <5 cm or three or fewer tumors individually <3 cm (Fig. 36.1). Using these selection criteria, patients transplanted had a favorable outcome, including a 4-year actuarial survival rate of 85% and a recurrence-free survival (RFS) rate of 92%. The suitability of these criteria for selecting patients for transplant has been confirmed by numerous studies (Table 36.6).^{73–82}

The excellent outcomes of HCC patients within the Milan criteria led many to explore more expansive and inclusive measures. The most accepted expanded criteria are from the University of California San Francisco (UCSF) group. They reported excellent results after transplant for solitary tumors ≤6.5 cm, three or fewer nodules with the largest ≤4.5 cm, and total tumor diameter ≤8 cm

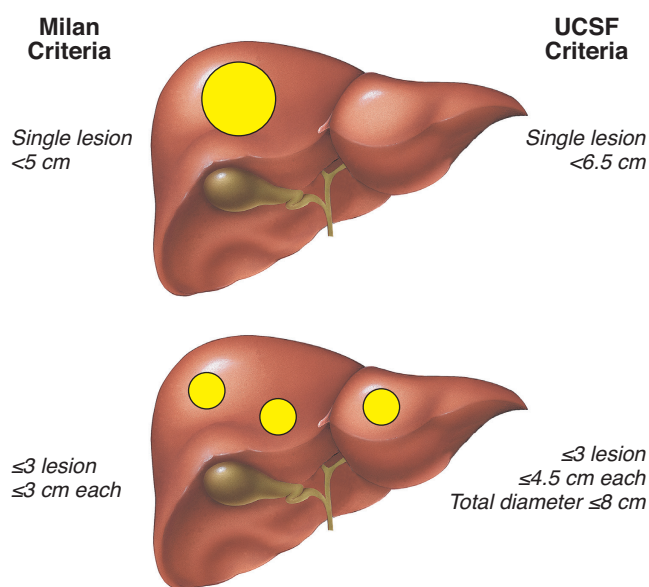


Figure 36.1 Milan (single lesion ≤5 cm, or no more than three lesions ≤3 cm) and University of California, San Francisco (UCSF) criteria (single lesion ≤6.5 cm, or no more than three lesions ≤4.5 cm and a total diameter of 8 cm) for patients with HCC and who are undergoing liver transplantation.

(see Fig. 36.1). The UCSF criteria were associated with 90% and 75% survival at 1 and 5 years, respectively.⁸³

The largest experience to date using transplantation for HCC was reported from the University of California, Los Angeles (UCLA).⁸⁴ In this study of 467 transplants performed for HCC, the overall 1-, 3-, and 5-year survivals were 82%, 65%, and 52%, respectively. Transplanted patients with tumors beyond the UCSF criteria had a 5-year survival below 50%.

TABLE 36.6

Results of Liver Transplantation for Early Hepatocellular Carcinoma

Study	Number of Patients	Tumor Number and Size	Neoadjuvant Therapy	Survival (%) ^a
Menahem et al., 2017 ⁷⁴	861	Small transplantable HCC or within Milan criteria	None	59
Grigorie et al., 2017 ⁷⁵	844	74% had single tumors, 26% had multiple	None	39
Hsueh et al., 2016 ⁷⁶	184	≤2.7 cm 55% had single tumors, 45% had multiple	None	76
Piros et al., 2015 ⁷⁷	49	Within Milan criteria (69.3%)	Immunosuppressive therapy with mTOR inhibitor	51
Mazzaferro et al., 2009 ⁸²	444	1 ≤5 cm; 3 ≤3 cm	None	73
	1,112	Beyond Milan criteria	None	54
Pelletier et al., 2009 ⁷⁸	2,790	1 ≤5 cm; 3 ≤3 cm	None	61
	346	Beyond Milan criteria	None	32
Herrero et al., 2008 ⁷⁹	47	1 ≤5 cm; 3 ≤3 cm	None	70
	26	Beyond Milan criteria	None	73
Onaca et al., 2007 ⁸⁰	631	1 ≤5 cm; 3 ≤3 cm	None	62 (5 y RFS)
	575	Beyond Milan criteria	None	43 (5 y RFS)
Yao et al., 2001 ⁸¹	46	1 ≤5 cm; 3 ≤3 cm	TACE/ETOH	72
Mazzaferro et al., 1996 ⁷³	48	1 ≤5 cm; 3 ≤3 cm	TACE	75 (4 y)

^aAll are 5-year survivals, except where noted. RFS, recurrence-free survival; ETOH, alcohol.

TABLE 36.7

Results of Living Donor Liver Transplantation for Hepatocellular Carcinoma with Patient Selection Based on the Milan Criteria

Study	Patients Meeting Milan Criteria		Patients Exceeding Milan Criteria	
	Number	3-Year Survival (%)	Number	3-Year Survival (%)
Sugawara et al., 2007 ⁸⁵	68	79	10	60
Canter et al., 2011 ⁸⁶	—	—	92	66
Gunay et al., 2015 ⁸⁷	57	86	8	81
Hong et al., 2016 ⁸⁸	364	88 (low) ^a 80 (medium) 60 (high)	168	80 (low) ^a 77 (medium) 9.1 (high)
Lee et al., 2017 ⁸⁹	181 (total) 47 in [GRWR <0.8%] ^b 134 in [GRWR ≥0.8%]	49 for [GRWR <0.8%] 68 for [GRWR ≥0.8%]	147 (total) 35 in [GRWR <0.8%] 112 in [GRWR ≥0.8%]	65 for [GRWR <0.8%] 80 for [GRWR ≥0.8%]
Lee et al., 2021 ⁹⁰	709	90	53	65

^a5-year disease-free survival rate for low, medium and high risk of recurrence based on AFP levels and PET scan results.

^bGRWR, graft-to-recipient body weight ratio; patients were stratified into two groups according to the GRWR (<0.8% and ≥0.8%).

Living Donor Liver Transplant for HCC

Because of the shortage of cadaveric livers, living donor liver transplant (LDLT) is increasingly used for treating patients with decompensated cirrhosis. In many Asian countries with high HCC prevalence, living-related transplants are the most common LTs performed. The disadvantage is the risk to the living donor, with morbidity as high as 40% and a 0.5% mortality. The greatest concern is that LDLT may encourage transplants for patients with unfavorable biology (outside of the established Milan or UCSF criteria) and pose an unjustified risk to two lives, including the healthy individual. Survival outcomes for all patients undergoing LDLT are favorable and, in most series, are comparable to the results with deceased donors (Table 36.7).^{85–90} Two studies summarize essential aspects of LDLT. Three thousand fifty-two patients from 12 collaborative centers in Europe, Asia, and the United States were examined, and LDLT decreased the intention-to-treat risk of death in HCC patients. A potential live donor prevented death while on the waitlist.⁹¹ In a meta-analysis involving 20 studies and over 9,000 patients, Yadav et al. found that salvage LTs had superior 1-, 3-, and 5-year OS rates than primary LTs in LDLT recipients.⁹² These data would indicate that patients with a resection option should have that as the primary treatment even if a salvage transplant may be needed later. The delayed time to transplant would allow for a better selection of patients more likely to benefit from transplantation. In cases where LDLT is necessary, such a strategy would also allow the risk to the donor to be better justified by the long-term survival of the recipient HCC patient.

Multimodality Management While Awaiting Transplant

To qualify for the waitlist, a biopsy or one of the following criteria must be fulfilled: AFP >500 ng/mL, arterial enhancement followed by a portal venous washout on computed tomography (CT) scans or magnetic resonance imaging (MRI), or a history of local-regional treatment. Patients should be assessed radiologically for the number and size of tumors to rule out extrahepatic disease and vascular involvement. Patients with tumors <2 cm or those who do not qualify for the Milan criteria can be listed for transplant, but they will receive no additional priority points for the tumor. The tumor should be assessed every 3 months by CT scans or MRI to rule out disease progression beyond the established criteria.

To reduce the likelihood of tumor progression while on the waitlist, many local treatments are used, including ADTs (trans arterial embolization [TAE]; trans arterial chemoembolization [TACE];

Yttrium-90 embolization [Y90]), percutaneous thermal ablation (radiofrequency ablation [RFA]; microwave ablation [MWA]), percutaneous ethanol injection (PEI), or stereotactic body radiation therapy (SBRT). For small, solitary tumors, PEI or RFA can be effective treatment options as a bridge to transplant. In a series of 52 patients treated with RFA, the dropout rate was 6% at 12 months due to tumor progression with a 3-year DFS of 76% for the 41 patients eventually transplanted.⁹³ Mazzaferro et al.⁹⁴ reported no dropout for patients treated with RFA as a bridge to transplant, with a 3-year survival of 83%. SBRT is increasingly used in this setting. A study from the University of Toronto compared outcomes of 379 patients treated with SBRT, ADT, and RFA as a bridge to transplant and found no difference in transplant dropout rate, postoperative complications, or survival, suggesting that this newer treatment modality could be a good alternative to more conventional therapies.⁹⁵

ADT, which involves selective embolization of the arterial feeding vessels for the hepatoma with occlusive particles with or without admixed chemotherapeutic agents, is often used for larger tumors. ADT can minimize waitlist dropout, decrease post-transplant recurrence, and downstage HCC beyond transplant criteria. A dropout rate of 14% was found in a series of patients treated with TACE as a bridge to transplant, which compared favorably to a dropout rate of 38% for an untreated group.⁹⁶ Recent results from a randomized phase II study suggest that SBRT may be equivalent to TACE as a bridge to transplant in HCC.⁹⁷

ADJUVANT AND NEOADJUVANT THERAPY

Adjuvant Therapy

There is no level 1 data supporting the use of adjuvant therapy after curative resection or ablation of HCC. Early encouraging data for adjuvant IFNα-2b were not substantiated by a subsequent more extensive study.⁹⁸ A recent meta-analysis involving 2,190 patients suggests that TACE following resection in patients with microvascular invasion may improve OS and DFS.⁹⁹

Sorafenib, discussed at length in the Systemic Therapy section, has been studied in the adjuvant setting. In a phase III trial (STORM), patients were randomized to adjuvant sorafenib versus placebo after surgery, RFA, or percutaneous alcohol injection with a primary end point of RFS. In this study, 1,114 HCC patients with a complete radiological response after surgical resection or local ablation were randomly assigned to receive sorafenib or placebo

TABLE 36.8**Recent Studies Comparing Long-Term Outcome of Patients with Hepatocellular Carcinoma Treated Primarily with Resection (and Salvage Transplantation) or Primary Liver Transplantation**

First Author	Year Published	Primary Therapy	Sample Size	5-Year Overall Survival Rate (%)	5-Year Disease-Free Survival Rate (%)
Menahem ⁷⁴	2017	Transplantation	861	60	62.5
		Resection	570	48	35.6
Shen ¹⁰⁴	2017	Transplantation	129	83	92
		Resection	209	67	48
Lim ¹⁰⁵	2012	Resection	4660 ^a	67	37
Lee ¹⁰⁶	2010	Transplantation	78	68	75 ^b
Lim ¹⁰⁵	2012	Resection	130	52	50
Facciuto ^{107c}	2009	Transplantation	119	62	—
		Resection	60	61	—
Del Gaudio ¹⁰⁸	2008	Transplantation	147	58	54
		Resection	80	66	41
Shah ¹⁰⁹	2007	Transplantation	140	64	78 ^b
		Resection	121	56	60
Poon ¹¹⁰	2007	Transplantation	85	44	—
		Resection	228	60	—

^aMeta-analysis.

^bSignificant difference as reported in the original study.

^cFourth-year survival rates are reported for patients meeting the Milan criteria.

Adapted from Rahbari NN, Mehrabi A, Mollberg NM, et al. Hepatocellular carcinoma: current management and perspectives for the future. *Ann Surg* 2011;253:453–469.

for 4 years. There was no difference in RFS (median 8.4 months with placebo vs 8.5 months with sorafenib) with HR 0.94.¹⁰⁰ A phase II randomized, placebo-controlled study evaluating whether adjuvant sorafenib can prevent HCC recurrence in high-risk OLT recipients is still enrolling (NCT01624285).

Brivanib, also discussed in greater detail in the Systemic Therapy section, is a selective dual inhibitor of VEGF and fibroblast growth factor (FGF) signaling. It was examined in the adjuvant setting following TACE in a large phase III study.⁶¹ Five hundred two patients were randomly assigned to brivanib (n = 249) or placebo (n = 253). Although OS did not improve (HR 0.90, 95% CI 0.66 to 1.23, *P* = 0.53), there did seem to be improvements in time to extrahepatic spread of disease (HR 0.64, 95% CI 0.45 to 0.9) and time to progression (TTP) (HR 0.61, 95% CI 0.48 to 0.77).⁶¹

Neoadjuvant Therapy

Two RCTs and seven nonrandomized trials (NRT) have evaluated preoperative transarterial chemotherapy. No clear advantage in DFS or OS was found in these studies.¹⁰¹

In summary, there is currently no clear role for neoadjuvant or adjuvant therapy in HCC treatment.

Treatment of Hepatitis C Before or After Treatment of Cancer

Mention must be made of the use of direct-acting antiviral agents in the treatment of hepatitis C. In 2014, Harvoni therapy was approved in the US to treat hepatitis C infection because data showed that a short course of treatment (2 to 3 months) could be curative.¹⁰² It is hoped that such antiviral treatments will decrease the need for LT and resection in the future.

Of note, there are emerging data that a small subset of patients being treated for hepatitis C after treatment for HCC may

experience a much more aggressive form of the cancer if recurrence occurs.¹⁰³ This observation is being verified by prospective ongoing studies, as well as the mechanisms are being studied.

Choice of Resection, Ablation, or Transplantation

Although liver resection versus liver transplantation as primary therapy for patients with small HCC and adequate hepatic reserve is hotly debated, in most cases, resection and OLT are complementary and not competing treatments. Several studies report comparable OS rates for primary resection and primary OLT for transplantable HCC (Table 36.8).^{74,104–110}

Patients with limited disease have potentially curative treatment options. There is a consensus that limited disease should be resected in patients with no underlying liver dysfunction as this provides the best chance of cure without an ongoing need for immunosuppression dictated by transplantation. The resection outcomes are quite good, with 5-year survivals of 30% to 50% (see Table 36.5).

Salvage Transplant

For patients with end-stage liver disease (ESLD) and limited HCC, OLT is currently the best treatment modality. However, OLT can only be offered to a small proportion of patients due to donor organ shortage. Thus, liver resection remains the most critical surgical therapy in HCC patients and compensated cirrhosis (CTP class A).⁵¹ Using resection as first-line therapy with salvage transplantation for recurrence is common.¹¹¹ Resection is essentially both a potentially curative cancer therapy and a bridge therapy before OLT. In a study from a large Asian transplant center, most patients (79%) who developed tumor recurrence after resection of small HCC were still eligible for salvage transplantation.¹¹² Primary resection also allows assessing the tumor and adjacent liver tissue pathologically. Pathologic prognostic factors

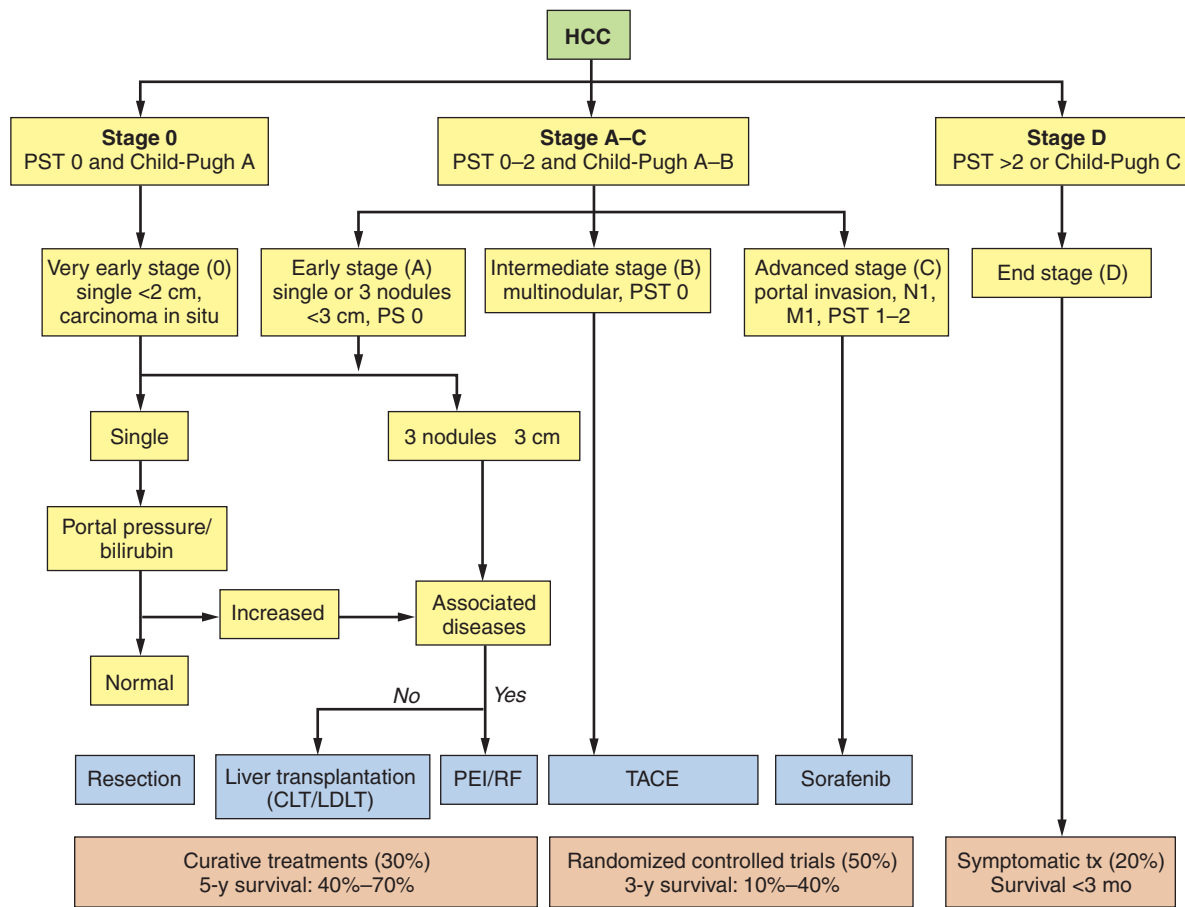


Figure 36.2 Strategy for staging and treatment assignment for patients with HCC according to the BCLC staging system. PST, performance status; CLT, cadaveric liver transplantation; tx, treatment.

such as micro- or macrovascular invasion, satellitosis, or occult tumors can be used as selection criteria for salvage LT (SLT) in the case of recurrent tumors.

A recent meta-analysis looked at the effectiveness and morbidity of SLT for initially resectable and transplantable HCC.⁹² PubMed, EMBASE, and the Cochrane Library were searched to identify studies comparing SLT and primary liver transplantation (PLT). SLT had superior 1-, 3-, and 5-year OS and DFS compared to PLT. In subset analysis for deceased-donor liver transplantation (DDLT) recipients, there were no significant differences in 1- and 3-year OS rates between SLT and PLT. However, for LDLT, the OS and DFS rates were superior in the SLT group compared to the PLT group. The meta-analysis suggests that SLT has comparable postoperative complications to PLT. Thus, SLT may be a better treatment strategy for recurrent HCC patients and patients with compensated liver, whenever feasible, considering the severe organ limitation and the safety of SLT.⁹²

Expanded Milan Criteria for Transplant

Patients with large HCC beyond the Milan or UCSF criteria have a less favorable outcome than those with small HCC. Patients in the US outside the accepted criteria can be transplanted based on the physiologic Model for ESLD (MELD) score but do not receive any exception points for HCC. In an analysis of 94 HCC patients exceeding the Milan criteria, resection results were compared to OLT.⁸⁶ The OS rate was 66% for both groups, even though the mean tumor size was 10 cm for the resection group and 6.4 cm for the OLT group, suggesting that resection and OLT in HCC patients beyond the Milan criteria have similar outcomes. A proposed algorithm of care from BCLC is shown in Figure 36.2.

Unfortunately, since the publication of the UCSF and BCLC expanded criteria for LT for HCC patients, many expanded criteria have been proposed, summarized in a recent review.¹¹³ This systematic review covering the period 1993 to 2021 identified 59 different inclusion criteria of HCC for LT. While the field works out a widely accepted “comprehensive” HCC-LT score to justify transplantation to more patients, the oncologist with a patient with few and small tumors should refer patients to the transplantation surgeon for evaluation for appropriateness of transplantation with local guidelines.

Locoregional therapies can downstage HCC, and in some cases, from beyond Milan criteria to within criteria. Investigators recently reported a phase 2b and 3 study evaluating the standard of care (SOC) treatment versus LT of patients who underwent downstaging with locoregional, surgical, and systemic therapies into Milan criteria. Forty-five patients were randomized, 23 to transplantation and 22 to the SOC, with significant patient outcomes in the transplant group. Specifically, the 5-year tumor event-free survival was 76.8% with transplantation versus 18.3% in the SOC group. Similarly, 5-year survival was 77.5% with transplant versus 31.2% with SOC, (HR 0.32, 95% CI 0.11 to 0.92, $P = 0.035$). This study expands the current LT criteria for liver-limited unresectable disease.¹¹⁴

Ablative Therapy for Localized Hepatocellular Cancer

Chemical Ablation

Chemicals destroy tumor tissue by direct dehydration of the cytoplasm, protein denaturation, consequent coagulation necrosis,

and indirect ischemia from vascular thrombosis from endothelial damage.^{115,116} The use of direct injection of chemicals such as absolute ethanol or acetic acid into tumors has been long studied for HCC treatment. Chemical ablation is relatively inexpensive and widely used in developing countries with a high HCC incidence. Moderate quality evidence supports that RFA is superior to chemical ablation for HCC treatment.¹¹⁷ Chemical ablation requires more applications and thus has been replaced by thermal ablation except in specific anatomic locations, such as adjacent to the major biliary tree, gallbladder, or bowel.¹¹⁸

Radiofrequency Ablation

Most studies on thermal ablation in the treatment of liver tumors use RFA. The term *radiofrequency* applies to all electromagnetic energy sources with frequencies <30 MHz, although most clinically available devices function in the 375 to 500 kHz range. The technique for thermal ablation in the liver using RFA was first described in the 1990s.¹¹⁹

In this technique, the RF electrode is placed into the tumor with imaging guidance or direct visualization in intraoperative ablation. The electrode is coupled to an RF generator and is grounded using a grounding pad or pads applied to the thighs. The RF generator produces a voltage between the active electrode (applicator) and the reference electrode (grounding pad), establishing electric field lines oscillating with the alternating current. This oscillating electric field causes electron collisions with the

adjacent molecules closest to the applicator, inducing frictional heating.¹¹⁹ Tissue heating to temperatures >60°C leads to immediate cell death. The goal of ablation is to create a zone of tissue necrosis that fully encompasses the tumor and a 5 to 10 mm margin of normal parenchyma. The volume of ablation achieved is based on the energy balance between heat conduction of the local RF energy applied and heat convection from the circulating blood and extracellular fluid.

Microwave Ablation

Like RFA, microwave ablation (MWA) uses electromagnetic waves to produce heating. Unlike RFA, the MW energy is not an electrical current and is in a much higher frequency range (300 MHz to 300 GHz). MW applicators available for clinical use generally operate in the 900 to 2,450 MHz range.¹²⁰ MW creates a much larger zone of active heating. The rapidly alternating electric field of the MW antenna causes water molecules to spin rapidly to align with electromagnetic charges of opposite polarity. MW tissue heating occurs because of kinetic energy induction in surrounding water molecules.

Currently, there are several commercially available MWA systems in the US and Europe¹²⁰ that use straight antennae with varying active tips 0.6 to 4.0 cm in length. Perfusion of the antenna shaft is required to reduce conductive heating of the nonactive portion of the antenna and prevent damage to tissues proximal to the active tip. Two systems can power up to three antennae with a single generator and treat large tumors (Fig. 36.3).

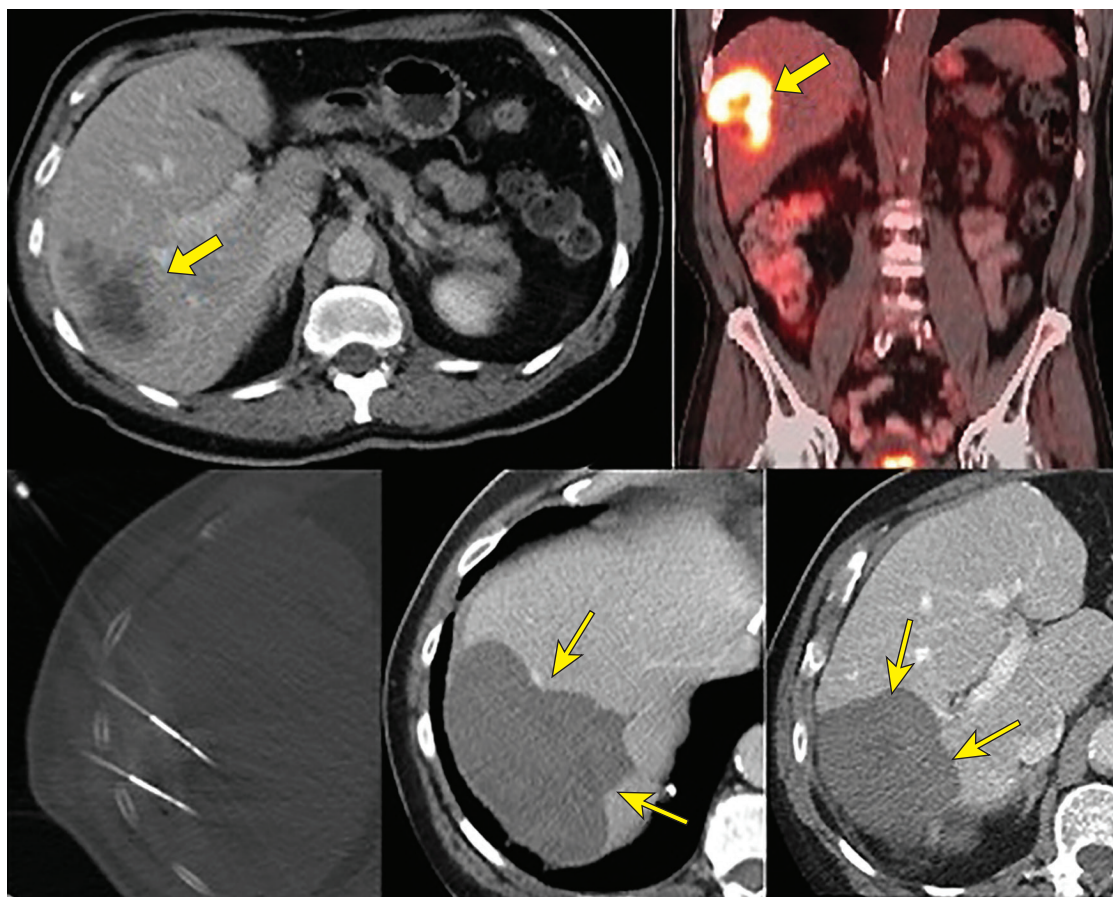


Figure 36.3 A 73-year-old man with previously resected colorectal metastatic disease in the left lobe of the liver recurred in the right lobe despite 5-fluorouracil, oxaliplatin, leukovorin (FOLFOX) chemotherapy. An axial contrast-enhanced CT scan and coronal fludeoxyglucose–positron-emission tomography (FDG-PET) CT scan (*top panel*) show a large heterogeneous colorectal cancer (CRC) metastasis in the right lobe. The patient refused additional chemotherapy, and ablation was offered as palliative therapy. Given the size of the mass, microwave ablation was performed with multiple antennae under CT guidance (*bottom left*). The patient responded to the treatment extremely well. A current 20-month follow-up contrast-enhanced CT scan (*bottom right*) shows a large coagulation defect (*small arrows*) without evidence of recurrence in the liver.

TABLE 36.9

Randomized Controlled Trials of Hepatic Resection Versus RF Ablations for HCC Reported After 2000

First Author/ Year Country	Treatment	Number	Age	Tumor Number	Characteristics Diameter	Disease-Free Survival (%)				Overall Survival (%)				p
						1 Year	3 Years	4 Years	5 Years	1 Year	3 Years	4 Years	5 Years	
Chen (2006) ¹²⁹	RES	90	49 ± 11	1	≤5 cm	87	69	46	—	93	73	64	—	NS
	RFA	71	52 ± 11	1	≤5 cm	86	64	52	—	96	71	68	—	
Liang (2008) ^{130,a}	RES	44	49 ± 12	≤3	≤5 cm	—	—	—	—	79	45	31	28	0.8
	RFA	66	55 ± 11	≤3	≤5 cm	—	—	—	—	77	49	40	40	
Huang (2010) ¹³¹	RES	115	56 ± 13	≤3	≤5 cm (77%)	85	61	—	51	98	92	—	76	NS
	RFA	115	57 ± 14	≤3	≤5 cm (73%)	82	46	—	29	87	70	—	55	
Feng (2012) ¹³²	RES	84	47(18–76)	≤2	≤4 cm	91	61	—	—	96	75	—	—	0.3
	RFA	84	51(24–83)	≤2	≤4 cm	86	50	—	—	93	67	—	—	
Wang (2014) ¹³³	RES	5,779	—	—	—	83.9	57.3	38.6	—	75.7	95.2	78.9	—	
	RFA	6,094	—	—	—	70.1	37.6	21.7	—	54.8	92.2	74.4	—	
Ng (2017) ¹³⁵	RES	109	55(31–82)	<3	2.9(1–5)	74.1	50.9	—	41.5	94.5	80.6	—	66.5	NS
	RFA	109	57(23–78)	<3	2.6(1–5)	70.6	46.6	—	33.6	95.4	82.3	—	66.4	

^aTreatments for recurrent disease.

RES, resected patients; OS, overall survival; DFS, disease-free survival; NS, not significant.

Irreversible Electroporation

Irreversible electroporation (IRE) is a relatively new, largely non-thermal electrical ablation technology. It is much less susceptible to the thermal heat sink that can limit the efficacy of thermal ablation devices in treating tumors adjacent to blood vessels and less likely to damage critical structures, such as major bile ducts.^{121,122} IRE uses high-power large electrical currents delivered between electrode pairs placed into tumors. Early treatment data in HCC have been published with small single-center series,¹²³ the largest in 58 patients that reported a one-year progression-free survival (PFS) of 70%.¹²¹

Outcomes of Ablations

Many factors affect the success of thermal ablation treatment for liver malignancies, some of which include tumor size, proximity to blood vessels, operator experience, presence of underlying liver disease, extrahepatic disease in patients with secondary liver malignancies, overall patient health, and the implementation of a multidisciplinary treatment clinic.

Tumors adjacent to larger (>3 mm) blood vessels may be undertreated due to the thermal sink effect.¹²⁴ Proper device selection to effectively eradicate tumors adjacent to vessels should be improved, in theory, with more active energy sources such as MWA compared to RFA. However, a randomized phase 2 trial in 152 patients comparing RFA and MWA for HCC up to 4 cm did not show a significant difference in local tumor progression (LTP) between modalities.¹²⁵ In clinical practice, less susceptibility to heat sink effects and faster treatment time, particularly with multiple simultaneous applicators, are attractive features of MWA compared to RFA.¹²⁶ A series of 459 HCC patients treated with MWA having actual 10-year follow-up reported survival at 10 years of 24% with 8.1% RFS.⁶²

Most HCC patients have underlying hepatitis or cirrhosis, which affects and sometimes limits treatment options and outcomes. HCC patients with CTP class C cirrhosis who are not on an LT waitlist and do not undergo liver-directed therapies have a median survival of <4 months. Therefore, treatment with thermal ablation is not warranted as it will not improve long-term survival.¹²⁷

In patients with CTP class A cirrhosis, however, outcomes of thermal ablation can rival surgery when tumors are solitary and smaller (<5 cm) and less in number (<3 tumors each under 3 cm). In retrospective studies, data suggest that for small HCCs (≤2 to 3 cm), RFAs result in similar outcomes as resections.¹²⁸

Several randomized trials comparing RFA and hepatic resection have been reported (Table 36.9).^{129–133} Chen et al. and Huang et al. compared tumors fulfilling the Milan tumor criteria for transplantation. Chen et al.¹²⁹ compared resection to RFA for tumors <5 cm. The 1-, 3-, and 4-year survivals were 93%, 73%, and 64%, respectively, for resection, and 96%, 71%, and 68%, respectively, for RFA. The authors concluded that RFA was as effective as surgical resection to treat solitary HCCs ≤5 cm in terms of OS and DFS after 4 years, with no significant difference in outcome between the two groups on follow-up. Based on a recurrence rate at 5 years of 63% in the RFA group and 41% in the resection group, Huang et al.¹³¹ concluded that surgical resections have better outcomes than RFAs. However, it must be noted that more patients in the resected group had tumors <3 cm. In addition, the OS was not statistically different between the two treatment groups.

In a recent meta-analysis involving 25 papers and over 5,000 patients, Shin compared outcomes of local ablation with that of resection.¹³⁴ Liver resection showed a significantly better 5-year RFS than RFA in the analysis of RCTs (HR: 0.75; 95% CI: 0.62 to 0.92; *P* = 0.006). The RFA group showed a significantly higher local recurrence than the liver resection group in both RCTs and NRTs. Additionally, the liver resection group showed better OS and RFS than the MWA or RFA plus TACE groups.

Feng et al.,¹³² Liang et al.,¹³⁰ and Ng et al.¹³⁵ confirmed similar efficacy of RFA to resection in tumors <4 cm. Both groups found RFA and resection to have similar OS in a follow-up period after 3 years. For small HCCs, particularly in patients with cirrhosis, RFA can produce similar cancer outcomes with much lower morbidity. A large meta-analysis evaluated 55 RCTs in 5,763 patients from 1988 to 2017 with preserved liver function and unresectable HCC comparing bland embolization, conventional chemoembolization, radioembolization (RAE) alone or in combination with thermal ablation, conventional chemotherapy, and external beam radiotherapy.¹³⁶ A propensity-score matching analysis of 154 patients that underwent TACE/RFA and surgical resection of

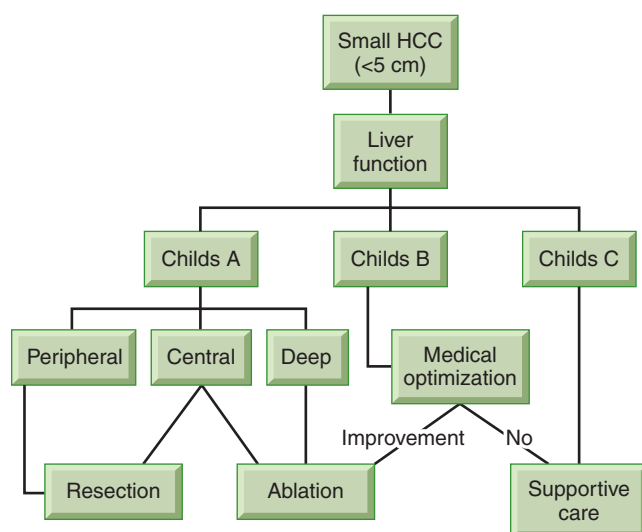


Figure 36.4 The treatment algorithm for a small HCC.

2 to 3 cm HCC found no significant difference in LTP, DFS, and OS between groups with LTP and intrahepatic distant recurrence rates of 12.9% and 34.3% versus 8.3% and 28.6% in the combined treatment group and surgical resection groups, respectively. In addition, the combination group had lower costs and fewer complications when compared to the surgical resection group.¹³⁷ Figure 36.4 illustrates a recommended algorithm of care for small HCCs.

Robotic Liver Resection and Evolving Issues on Use of Resection and Ablation

There is no doubt that both resection and ablation can cure patients with small HCC. They are complementary therapies. In patients after either resection or ablation, the other therapy may be chosen to treat recurrences when they happen. In patients with multifocal disease, a combination of resection and ablation may be chosen to minimize parenchymal loss, which optimizes the eradication of cancer. Thus, it is rarely a matter of choosing one therapy over the other, but rather a choice of the order of therapies.

Recent developments in robotic surgery have further enhanced minimally invasive therapies for HCC. Liver resections can now be performed robotically as effective treatments for liver cancers with minimal morbidity, often with very short hospital stays, including outpatient procedures.¹³⁸ Some investigators have compared robotic resection to ablation.¹³⁹ Magistri et al. examined outcomes for their robotic liver resections (RLR) and percutaneous ablation (PA) between 2014 and 2019 for newly diagnosed single, small (<3 cm) HCC to compare short- and long-term outcomes. In total, 24 RLR and 36 PA were performed. There were no statistically significant differences regarding the incidence of postoperative morbidity (RLR 38% vs PA 19%, $P = 0.15$). The cumulative incidence of recurrence (CIR) was significantly higher in patients who underwent PA, with the 1, 2, and 3 years of CIR being 42%, 69%, and 73% in the PA group, and 17%, 27%, and 27% in the RLR group. Though this study indicates RLR to have more durable responses, in reality choice between RLR and PA usually is dictated by anatomic location. For small peripheral lesions, RLR is technically easy and usually results in an outpatient encounter. For deeper lesions, PA offers the opportunity to save more parenchyma. Both are highly effective as cancer therapies.

Combination ADT and Ablation

The combination of ADT and ablation has been reported for HCC up to 7 cm in size. There are hypothetical advantages to the combination of the two locoregional therapies. For example,

the devascularization of the tumor can enhance the volume of ablation because of a decreased heat sink, and ADTs can treat micrometastases beyond the ablation margin. A retrospective study that followed 73 patients with solitary HCC up to 7 cm for a median of 134 months found no significant difference in OS or intrahepatic disease progression between the 40 that underwent surgical resection and 33 combination bland embolization and ablation. However, resection patients were less likely to require salvage treatments.¹⁴⁰ In two prospective studies with 318 patients, Peng et al. compared sequential TACE plus RFA versus RFA alone in HCC patients. For tumors >3 and up to 5 cm, combination TACE plus ablation improved OS in patients with newly diagnosed and recurrent HCC.^{141,142}

Tumors >5 cm are not optimal candidates for ablation and may be better treated with resection, when feasible, or arterially directed therapies including bland embolization, chemoembolization, or RAE. In a randomized study of 240 patients with a solitary recurrent HCC <5 cm, recurrent hepatectomy improved oncologic outcomes compared with percutaneous RFA. This was specifically true for tumors >3 cm (HR 1.72, 95% CI 1.05 to 2.84) or an AFP level >200 ng/ml (HR 1.85, 95% CI 1.15 to 2.96), reinforcing the benefit of resection of larger tumors if feasible in surgical candidates.¹⁴³

Arterially Directed Therapies for Regional Disease

AASLD, EASL, and consensus guidelines recommend arterially directed therapy (ADT) for patients with BCLC B (intermediate stage) disease. Intermediate stage disease includes multifocal liver-confined tumor in patients with preserved liver function and good performance status (ECOG 0). In practice, the indications for ADT often extend to include patients with BCLC C (advanced stage) that have limited extrahepatic disease and/or portal vein tumor involvement or mildly compromised performance status. These techniques rely on the dual blood supply of the liver. Approximately 75% of the nutrient blood flow to the liver comes from the portal vein, and most of the tumor supply is from the hepatic artery. Therefore, we can administer intraarterial treatment safely to the tumor with little effect on the non-tumor-bearing liver parenchyma.

There are three primary percutaneously administered transcatheter intra-arterial therapies: bland embolization, TACE, and RAE. Notably, while each technique has a slightly different complication profile, none is superior to another in OS. Therefore, the method used in a given patient is subject to local expertise and patient-related factors.

Bland embolization refers to the use of small particles ($\leq 100 \mu$) to induce complete stasis of the arterial supply to the tumor to cause ischemic necrosis. On the other hand, TACE involves an intra-arterial injection of chemotherapeutic agents, typically mitomycin C and/or doxorubicin followed by embolization of the feeding arteries to prevent washout of the chemotherapeutic agents, thereby aiming to achieve higher intratumoral drug concentration. In both cases, it is likely that the majority of the effect results from tumor ischemia induced by occlusion of the arterial vessels. RAE, on the other hand, is somewhat of a misnomer because it does not embolize the tumor supply but involves the local administration of yttrium-90 (Y90, a pure β emitter) loaded in glass or resin microspheres.¹⁴⁴

Bland Embolization

Performance status, liver function, and degree of portal hypertension are important patient selection criteria. Evaluation in a multidisciplinary team including hepatobiliary surgeons, gastroenterologists, medical oncologists, radiation oncologists, and surgeons has been shown to improve outcomes and direct more patients to treatment.¹⁴⁵ Most patients experience a postembolization syndrome of pain, fever, and nausea that may last several days to a few weeks.

In 2002, two studies demonstrated the benefit of TACE compared to best supportive care.¹⁴⁶ One of these trials was performed in Europe, randomizing 112 patients into three arms: conservative treatment, bland embolization with gelfoam, and doxorubicin TACE. Sequential analysis showed improved survival for TACE compared to conservative treatment. Unfortunately, the trial was stopped before any statement about the utility of bland embolization (with gelfoam, a precursor to the well-calibrated agents in current use) compared to either conservative treatment or TACE. These studies led to the adoption of TACE as the SOC in many institutions. There has been no further prospective trial comparing bland and conventional TACE.

The data on TACE are limited by the number of variables in how the procedure is performed, including chemotherapeutic agents (dose, number, and type), embolic agents (gelfoam vs calibrated, size), level of embolization (subselective, segmental, lobar), and schedule (on-demand vs regularly scheduled).

In the early 2000s, drug-eluting embolic beads (DEB) were introduced to overcome some of the limitations of conventional TACE. Chemotherapy, including doxorubicin, can be loaded onto the beads by disulfide bonds and is eluted over time once deposited in the tumor. An early prospective study comparing bland embolization and doxorubicin DEB-TACE in 84 patients found improvement in TTP with DEB-TACE (42 vs 36 weeks) but no survival benefit. This study used particles 100 μ to 300 μ and 300 μ to 500 μ in size, larger than typically used for bland embolization, particularly as smaller and more well-calibrated particles are known to improve delivery into the tumor bed.¹⁴⁷

A more recent randomized phase II study compared bland embolization and doxorubicin DEB-TACE (both arms treated with 100 μ to 300 μ particles) and found no significant difference in RECIST response, PFS, or OS between treatment arms. One hundred one patients were randomly assigned to bland embolization or DEB-TACE. Although there was a nonsignificant improvement in PFS with DEB-TACE (6.2 months vs 2.8 months, $P = 0.11$), there was no difference in OS (19.6 vs 20.8 months, HR 1.11, $P = 0.64$). This study challenges the routine use of doxorubicin-eluting beads for chemoembolization.¹⁴⁸

Subsequently, a meta-analysis of RCTs evaluating bland embolization, TACE, DEB-TACE, and RAE evaluated 55 trials with 5,763 patients with BCLC stage B-C HCC. Each of the ADTs had a significant improvement in OS but no significant difference in objective response or OS.¹³⁶

Transarterial Chemoembolization

As mentioned earlier, there is a level I evidence of a survival benefit to conventional TACE compared to conservative management.^{146,149} In the trial by Lo et al.,¹⁴⁹ predominantly Asian patients with hepatitis B were randomized to TACE (Cisplatin + Lipiodol + Gelfoam) versus control (no treatment). The 2-year survival was 31% for TACE versus 11% for controls. In the trial by Llovet et al.,¹⁴⁶ predominantly European patients with hepatitis C or alcoholic cirrhosis were randomized to TACE (Doxorubicin + Lipiodol + Gelfoam). They had a 2-year survival of 63% versus 27% for control.

TACE with Sorafenib Versus Sorafenib Alone in Advanced Unresectable HCC: A phase III study compared the safety and efficacy of selective chemoembolization (TACE) combined with sorafenib to sorafenib alone in similar patients with locally advanced, unresectable HCC. In this study, 339 patients were randomly assigned to sorafenib + TACE ($n = 170$) versus sorafenib alone ($n = 169$). There was no significant improvement in OS with the addition of TACE, with a median OS of 12.8 months with sorafenib + TACE and 10.8 months with sorafenib alone.¹⁵⁰ There was an improvement in TTP, PFS, and tumor response rate (RR). Combination studies of TACE with newer systemic therapies have been initiated.

Brachytherapy/Radioembolization

Radioembolization (RAE) or selective internal radiation therapy involves the administration of glass (TheraSphere, Boston Scientific Marlborough, MA, USA) or resin (SIR Sphere, Sirtex Medical, Australia) beads loaded with Y90 into the arterial supply of the tumor. Y90 is a pure beta emitter with a range of only a few millimeters in tissue and a half-life of 2.67 days. Standard dosimetry is calculated based on the volume of the liver to be treated and the intended dose to the tumor, assuming uniform distribution in the territory treated. A pretreatment angiography is required to “map” the treatment area, calculate dose and evaluate for the presence of nontarget vessels and lung shunt fraction to ensure safe administration.

Radiation segmentectomy and radiation lobectomy are emerging techniques that use ablative doses of radiation to improve tumoricidal effect and cause contralateral hepatic hypertrophy, respectively. In radiation segmentectomy, a tumor localized to a single segment can be selectively treated with a dose calculated for the entire liver lobe, achieving doses of over 400 Gy to cause segmental liver ablation. The explanted livers of 33 patients with HCC <5 cm who underwent an LT after radiation segmentectomy with a median of 242 Gy demonstrated 90% to 100% necrosis in all tumors.¹⁵¹

Radiation lobectomy is performed to treat tumors and induce ipsilateral lobar atrophy and contralateral hypertrophy in patients who are potential candidates for resection. A modified approach of this technique is to administer a relatively high dose (>190 Gy) to the tumor-bearing segment followed by an ipsilateral lobar dose of 80 to 120 Gy to promote contralateral hypertrophy.⁴⁸

RAE does not improve survival compared to bland embolization or TACE, but some data suggest an improved TTP compared to TACE. The PREMIERE trial¹⁵² was a prospective phase II study that randomized 45 BCLC A/B HCC patients to RE versus TACE. Patients in the RE arm had a significantly longer TTP (median >26 months) than 6.8 in the TACE arm, but, again, no difference in OS was reported.

Because it is not an embolic therapy, Y90 is the preferred ADT in patients with portal vein tumor thrombus (PVTT), although bland embolization and TACE have also been used successfully. However, two phase III studies comparing the safety and efficacy of selective RAE to sorafenib in locally advanced BCLC C, unresectable HCC patients did not show improved OS compared to sorafenib. Chow et al. randomized 360 patients, with a median OS of 8.8 months with selective RAE and 10.0 months with sorafenib. Notably, RAE was associated with significantly fewer serious adverse events, 20.8% versus 35.2% with sorafenib, suggesting that RAE is viable for some locally advanced HCC patients.¹⁵³ The SARAH trial was a multicenter study that randomized 467 patients, 237 to Y90 and 222 to sorafenib. Fifty-three of the 237 (22%) randomized to Y90 did not receive Y90 and were treated with sorafenib. With a median follow-up of 27.9 months, there was no difference in OS (8 months in Y90, 9.9 in sorafenib HR 1.15 with 95% CI 0.94 to 1.41).¹⁵⁴

Side effects of RAE consist of mild nausea, pain (although typically less than TACE), and fatigue. Rare complications include radiation-induced liver disease, radiation pneumonitis, and GI ulceration from nontarget embolization. Risk factors associated with early mortality include an infiltrative tumor, a tumor encompassing over 50% to 70% of the liver, albumin <3 g/dL, bilirubin >2 mg/dL, and lung dose >30 Gy.¹⁵⁵

Radiation Therapy as Local Ablation for Hepatocellular Carcinoma

Radiation therapy (RT) for liver tumors was historically limited by hepatic toxicity. With improved imaging, treatment planning and delivery, and image guidance, it is now an excellent option, particularly for patients with unresectable tumors or who are medically

inoperable. Tumors not likely to be effectively treated by RFA due to size >3 cm or proximity to the diaphragm, large vessel, or gallbladder are also good candidates for RT. With proper care, tumors near gastrointestinal (GI) structures can also be treated with RT. Side effects are typically minimal and include mild fatigue, and less commonly nausea, mild radiation dermatitis, tumor edema-associated pain, or worsening liver dysfunction. Patients with metastases to the bone, brain, adrenals, or other locations can be effectively palliated with RT, as can patients with pain from large primary tumors.¹⁵⁶ Treatment-related toxicities are generally very low, particularly with SBRT, with minimal impact on quality of life demonstrated in a 222 patient prospective longitudinal study from the University of Toronto.¹⁵⁷

External Beam Radiotherapy

Fractionated Treatment

Because the liver is one of the most radiosensitive organs in the body, treatment planning and delivery must be done carefully to maximize the dose to the tumor and minimize it to the normal liver. The primary toxicity of concern is a radiation-induced liver disease (RILD), categorized as classic and nonclassic. Classic RILD is a constellation of anicteric ascites, hepatomegaly, and elevated liver enzymes (particularly alkaline phosphatase), which typically occurs within 4 months of therapy and is a type of veno-occlusive disease, similar to that seen after high-dose chemotherapy conditioning for bone marrow transplantation.¹⁵⁸ Nonclassic RILD, a more recent classification, is characterized by jaundice and markedly elevated serum transaminases (>5 times the upper limit of normal) within 3 months of therapy completion. This is thought to represent direct hepatocyte rather than endothelial injury.¹⁵⁹ RILD is typically self-limited but can be severe, even leading to mortality. It is managed symptomatically using diuretics and paracentesis. Even low doses to the whole liver of 25 Gy in 10 fractions or 32 Gy in 1.5 Gy per fraction twice daily are associated with a >5% risk of RILD, particularly for patients with cirrhosis and already compromised liver function. Other toxicities that can occur include gastric or duodenal bleeding,¹⁶⁰ although both of these risks can be minimized using careful treatment planning, image guidance, and treatment delivery. It is essential to consider factors including cirrhosis, prior liver-directed therapies, and hepatitis B and C, which add to the dose and volume models for the prediction of RILD. ICG is also used in Asia as a pretreatment assessment for the safety of RT, similar to its use for the safety of liver resection.¹⁶¹ In the US, the University of Michigan has pioneered its use to measure individual tolerance to radiation, which can be quite variable. Even some patients with good pretreatment liver function can experience rapid decline, so customizing treatment is crucial.¹⁶²

At the University of Michigan, investigators created the normal-tissue complication probability (NTCP) model that quantitatively described the relationship between radiation dose and liver volumes and the probability of developing RILD.¹⁶³ The dose was customized per patient to give an anticipated risk of 5% to 15%. Due to the variety of tumor and liver sizes and geometries, prescribed doses ranged from 40 Gy to 90 Gy. Patients who received 75 Gy or more had a higher median survival of 24 versus 15 months.¹⁶⁴ A prospective phase II trial from France treated patients to 66 Gy in 2 Gy fractions, with a RR of 92%.¹⁶⁵ In a multicenter retrospective patterns-of-care study among 10 institutions in Korea, 398 HCC patients were described. Of those, 70% were CTP class A, 54% had tumors >5 cm, and 40% had PVT. Nearly all had received prior treatment, mostly TACE. In this series, a higher dose (effective biologic dose over 53 Gy) correlated with better survival.¹⁶⁶ Other groups have combined TACE and RT with various schedules, although mostly using RT to treat residual disease. In a Korean study of 73 patients with incomplete responses to TACE, 38 received RT, whereas the rest received additional

TACE. Patients who received radiation had a higher 2-year survival of 37% versus 14% ($P = 0.001$).¹⁶⁷

Another use for RT is to treat PVTT to decrease portal pressure and allow the safe delivery of embolic therapies. The largest series from Taiwan reports a 25% RR. Compared with a dismal 1-year survival of 9% for nonresponders, responders had a better survival of 21% for those still not eligible for definitive therapies and 29% for those who ultimately had additional therapy.¹⁶⁸ A recent trial in 90 HCC patients and macroscopic vascular invasion randomized patients to sorafenib alone or a combination of TACE + RT. The primary end point was short, but at 12 weeks, PFS was higher in the TACE-RT group (87% vs 34%). The radiographic RR (33% vs 2%) and OS were also higher with TACE-RT.¹⁶⁹

Other crucial components of a liver RT program not always available for standard RT include arterial and portal venous phase imaging with CT scans and MRI for tumor delineation, motion assessment, and management tools, including 4D-CT scans and treatment gating, and precise image guidance (CT scan or MRI) on the treatment machine. Because liver tumors move with respiration, accurate assessment and management will aid in proper tumor targeting and normal tissue avoidance. Standard methods include forced or voluntary breath-hold for tumor immobilization and 4D-CT scans to assess and help account for motion.¹⁶³ Fiducials can also be placed within or near the tumor percutaneously before treatment planning, with daily alignment on the treatment machine, because external anatomy is a poor surrogate for internal anatomy and tumor location. However, cone beam CT scans often provide adequate image guidance without fiducials.

Stereotactic Body Radiotherapy

Stereotactic body radiotherapy (SBRT) is a relatively new method of delivering high-dose, high-precision therapy in a few treatments rather than daily over several weeks. With high doses per fraction, the biologic effect is greater than with the same dose delivered in a standard fractionated course, up to the equivalent of 80 to 150 Gy in 2-Gy fractions (Fig. 36.5). This technique was pioneered by Blomgren et al.,¹⁷⁰ who treated 20 liver tumors, including 8 HCCs, in the 1990s. Since then, the literature has been populated by small retrospective studies and prospective studies. Mendez-Romero et al.¹⁷¹ initially started treating patients with a schedule of 25 Gy in five fractions, but after two patients experienced local failure, the dose was raised to 30 Gy in three fractions for the rest of the study. The 1-year local control was 75%, with all failures in the low-dose group. One CTP class B patient experienced grade 5 nonclassic RILD, leading the authors to advise others to be cautious with similar patients.¹⁷¹ The subsequent and most extensive studies were a phase I and II trial from Princess Margaret Hospital.¹⁷² Of 41 initial patients, 31 had HCC. Patients were treated with a 6-fraction course of SBRT delivered over 2 weeks, with the dose individualized based on the University of Michigan NTCP model, escalated from 5% to 20%. The median dose was 36 Gy (range, 24 Gy to 54 Gy), and the median tumor volume was 173 mL (range, 9 to 1,913 mL). Even for these relatively large tumors, ~50% responded to treatment, and 42% had stable disease. Median survival was 12 months. No patients developed classic RILD, although 26% had grade 3 elevation of liver enzymes and 16% declined from CTP class A to B within 3 months after treatment. In the entire cohort of 102 patients from both the phase I and II portions of the trial,¹⁷² 93% had underlying liver disease, 52% had received prior therapy, and 55% had tumor vascular thrombosis. The 1-year local control was excellent, at 87%. However, 30% of patients had grade 3 or higher toxicity, possibly contributing to mortality. This illustrates the tenuous balance between treatment safety and efficacy and the need for improved predictive safety models for individual patients.

Another phase I study reported 100% local control after 36 to 48 Gy in three to five fractions. Similar to the Mendez-Romero study, patients with CTP class B liver failure were prone to liver toxicity.¹⁷³ A small prospective study of 29 patients with CTP B or C HCC used a median dose of 30 Gy in six fractions. Sixty-three

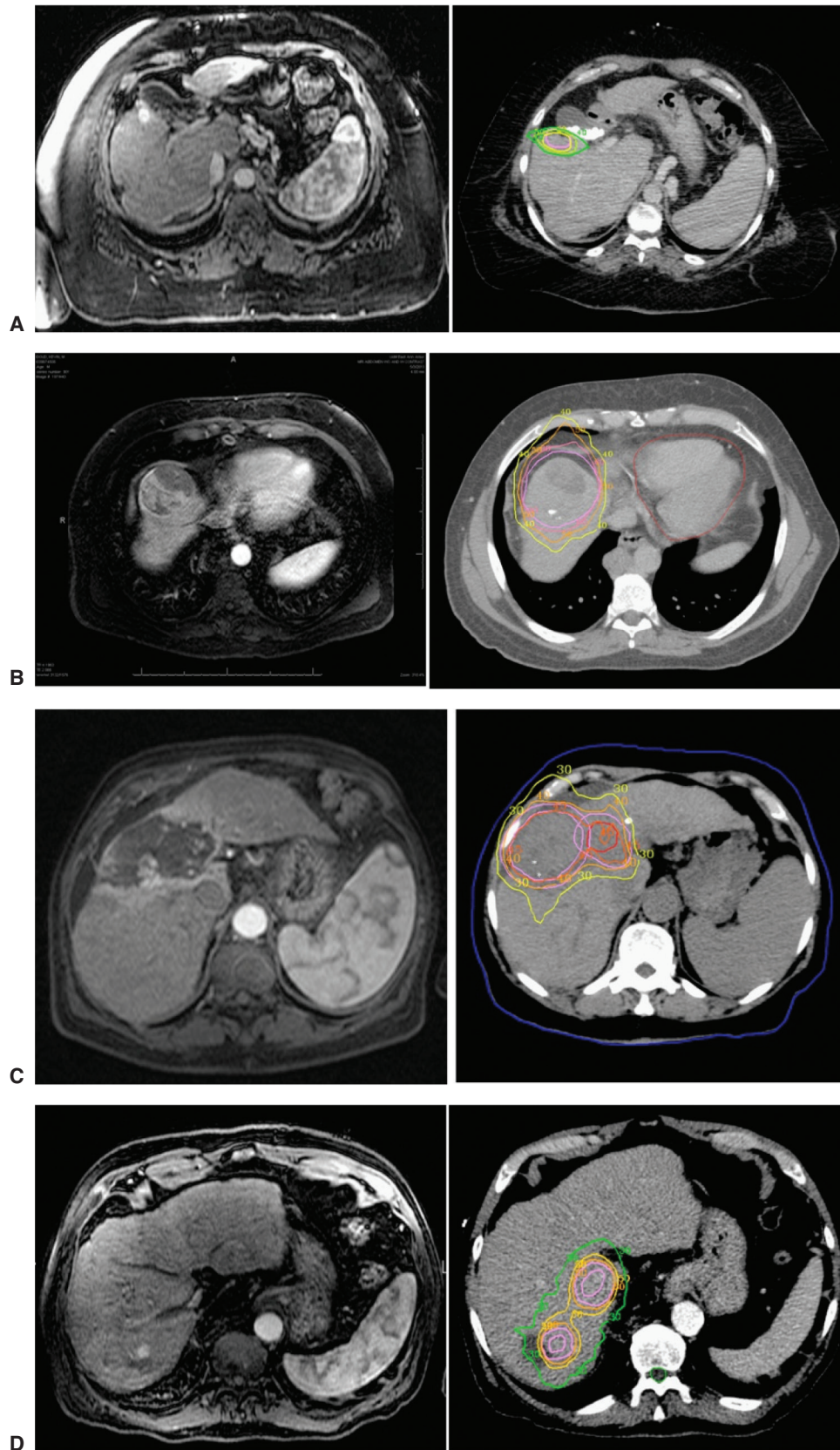


Figure 36.5 An SBRT can be used to treat single (**A, B**) or multiple (**C, D**) tumors in three to five high-dose, highly focused treatments. Planning is typically performed by fusing MRI images (**A, C**) with a planning CT scan (**B, D**) for accurate targeting. Special attention must be given to minimizing radiation dose to the remaining liver, particularly for patients with preexisting liver dysfunction.

percent of patients had a decline in CTP score by at least two points at 3 months.¹⁷⁴ Indeed, these patients need extra attention and consideration. Many larger retrospective studies have confirmed these smaller prospective results.¹⁷⁵ Rather than choosing efficacy or safety, a new strategy of individualizing and adapting therapy for patients to maximize *both* the safety and efficacy of treatment was recently reported for patients at high risk for liver damage.¹⁷⁶ Ninety patients with intrahepatic cancers received SBRT, with the dose guided by each individual's tolerance to liver radiation as measured by a change in ICG clearance during therapy. Treatment was adjusted using a Bayesian adaptive model to maintain safety while optimizing tumor control. The 2-year local control was 95%, with only 7% of patients experiencing a two-point decline in CTP score at 6 months. In the future, additional biomarker-based treatment personalization will likely become more common.

When deciding between ablation and SBRT, tumor and patient features may help guide clinicians. In a 2016 comparison of over 300 tumors treated with the two modalities, there was no difference in local control for tumors <2 cm in greatest dimension, but for tumors ≥ 2 cm, there was decreased freedom from local progression for RFA compared with SBRT.¹⁷⁷ This difference was significantly pronounced with increasing tumor size. The grade 3+ complication rate was similarly low for both treatments. This has subsequently been confirmed in a larger multi-institutional retrospective study in Asia. A 2020 meta-analysis found equivalent local control (LC).¹⁷⁸ Other factors influencing the treatment decision could be anesthesia tolerance, invasiveness of RFA, and ability to travel for three to five treatments of SBRT.¹⁷⁹

Charged Particle Radiotherapy

Charged particles, including protons and carbon ions, are being investigated for HCC treatment. These have the advantage of lower entrance and minimal exit dose due to a difference in dose deposition properties. The most extensive experience is from Tsukuba University, in which 318 patients were treated with 72 Gy in 16 fractions using protons with a 5-year local control and survival of 83% and 45%, respectively. There were no grade 3 or higher toxicities.¹⁸⁰ In a phase II proton study, 51 patients received 66 Gy in 10 fractions. The 5-year LC and survival rates were 88% and 39%, respectively. There were no cases of RILD, although 26% had a decline in liver function. A phase II multi-institutional study from the US was reported, in which 92 patients with HCC or intrahepatic cholangiocarcinoma received 15 fraction proton therapy to 40.5 to 67.5 GyRBE, based on limiting the risk of liver damage.¹⁸¹ Two-year local control for HCC patients was a promising 95%, with a 63% 2-year OS and only 5% grade 3 toxicity. The next few years should yield additional exciting developments, including guidance on patient/tumor selection for proton versus photon RT.

Currently, carbon-ion therapy is only available in a few centers worldwide. In Japan, 124 HCC patients have been treated on a phase I/II dose-escalation study of hypofractionated carbon-ion RT. They received from 50 Gy to 80 Gy in 15 fractions, with 3-year LC and survival of 91% and 50%, respectively.¹⁸² Despite the technologic imperative that newer is better, charged particles are not necessarily the miracle treatment some may assume. The high-dose treatment region cannot always be shaped as conformally as with standard photon radiotherapy (SBRT and intensity-modulated radiotherapy [IMRT]), and image guidance is relatively rudimentary with most systems. Therefore, ensuring the treatment accuracy is difficult. However, because the liver is extremely sensitive to radiation, reducing the moderate and low-dose regions could aid in protecting the normal liver and allowing for escalation of tumor dose. Thus, research into charged particle therapy is ongoing.

Toxicities

RT is typically very well tolerated. Mild fatigue can be due to the treatment itself and travel related due to multiple appointments

for the treatment. Mild nausea can be prevented by premedicating with an antiemetic. Radiation dermatitis is highly unusual due to the highly conformal radiation dose distribution. Occasionally, the treatment of large tumors can cause a pain flare due to local edema. These acute effects typically resolve within a few weeks after treatment. Late effects could include RILD, as discussed previously, in addition to GI ulceration and renal dysfunction. However, these risks can be avoided by minimizing the dose to these structures to below acceptable levels.¹⁶⁰

Systemic Therapy for Hepatocellular Carcinoma

First-Line Single-Agent Therapies

For HCC, many controlled and uncontrolled clinical studies have been performed with major classes of cancer chemotherapy, given intravenously as a single agent or in combination. Disappointingly, systemic chemotherapies have had no proven benefits on survival in HCC.¹⁸³ Many other nonchemotherapeutic agents have been tried without definitive success, including luteinizing hormone-releasing hormone agonists, tamoxifen, IFN, Sandostatin, megestrol, vitamin K, thalidomide, interleukin-2, ¹³¹I-lipiodol, and ¹³¹I-ferritin.

Given the vascular nature of HCC, that VEGF promotes HCC development and metastasis, and that increased levels of VEGF are associated with inferior survival,¹⁸⁴ antiangiogenic agents have been studied extensively in the setting of advanced HCC. The effect of sorafenib, a multityrosine kinase inhibitor with antiangiogenic effects, is thought to be mediated by the blockade of VEGFR-2/-3, platelet-derived growth factor receptor (PDGFR)- β , and other receptor tyrosine kinases.¹⁸⁵ The clinical efficacy of sorafenib in HCC was first reported in a multicenter phase II study of 137 patients with systemic treatment-naïve, inoperable HCC and varying hepatic reserve (72% CTP class A, 28% CTP class B).¹⁸⁶ Although only 2.2% of the study population achieved a confirmed objective response by WHO criteria, 42% had extended disease control. There was a median OS of 9.2 months, which was encouraging compared to historical controls.

Subsequently, the Sorafenib Hepatocellular Carcinoma Assessment Randomized Protocol (SHARP) trial enrolled 602 patients (majority recruited from the Western Hemisphere) with advanced HCC and CTP class A status who had not received prior systemic therapy.¹⁸⁷ Patients were randomly assigned to receive sorafenib at 400 mg orally twice a day (n = 299) or best supportive care (n = 303). The co-primary end points of the study were OS and time to symptomatic progression. Median OS was 10.7 months in the sorafenib arm versus 7.9 months in the placebo arm (HR = 0.69; 95% CI, 0.55 to 0.87). A predefined subset analysis indicated that the survival benefit of sorafenib was independent of performance status and disease burden.

Parallel to the SHARP trial, the Asia-Pacific study assessed the efficacy and tolerability of sorafenib in comparison with best supportive care in advanced HCC patients localized to Asia.¹⁸⁸ As expected, and in contrast to the SHARP trial, the Asia-Pacific study was enriched with HBV-related HCC patients (73% of the total study population). The trial confirmed that sorafenib, compared to best supportive care, was tolerable and led to a statistically significant improvement in disease control, time to radiographic progression, and OS. However, the magnitude of the OS benefit on the Asia-Pacific study was not as substantial as that observed in the SHARP study—the median OS was only 6.5 and 4.2 months for patients receiving sorafenib and placebo, respectively. Potentially, the inclusion of a larger percentage of patients with extrahepatic spread HCC (68.7% in Asia-Pacific vs 53% in SHARP) before beginning therapy might have accounted for differences seen in median OS between the Asia-Pacific and SHARP studies.

Sorafenib with or without TACE in Hepatocellular Carcinoma

In addition to single-agent sorafenib, investigators also examined the role of TACE to improve the activity of sorafenib in a randomized phase III study. Three hundred thirty-eight advanced HCC patients were randomly assigned to receive sorafenib alone ($n = 169$) or sorafenib plus TACE ($n = 169$). The median OS was 12.8 months with sorafenib + TACE versus 10.8 months with sorafenib alone, HR = 0.91; 95% CI 0.69 to 1.21; $P = 0.29$. These data confirm that TACE does not improve the efficacy of sorafenib alone in advanced HCC.¹⁵⁰

Since the approval of sorafenib, several other small receptor tyrosine kinase inhibitors have been studied. For the most part, results have been disappointing, with several major phase III studies of antiangiogenic therapy failing to improve survival outcomes versus sorafenib in the first-line setting. Sunitinib, an inhibitor of VEGFR-1/-2 with greater potency than sorafenib, plus an inhibitor of PDGFR- α/β , c-KIT, FLT3, RET, and other kinases, had shown anticancer activity in phase II studies.¹⁸⁹ However, in a subsequent randomized phase III study in CTP class A advanced HCC patients, the median OS for the sunitinib arm was 7.9 versus 10.2 months for the sorafenib arm (HR = 1.30; one-sided $P = 0.9990$; two-sided $P = 0.0014$).

Brivanib demonstrated modest antitumor activity in both treatment-naïve patients and those patients who had failed prior antiangiogenic therapy in phase II studies.¹⁹⁰ In a large randomized phase III study comparing brivanib to sorafenib in treatment-naïve advanced HCC patients,¹⁹¹ the median OS with brivanib treatment was 9.5 months versus 9.9 months with sorafenib (HR = 1.06; 95% CI, 0.93 to 1.22; $P = 0.37$).

Linifanib, a selective inhibitor of VEGFR and PDGFR, also failed to improve upon the modest survival advantage of sorafenib in phase III studies, despite early encouraging efficacy data.¹⁹² Bevacizumab, a monoclonal antibody directed against VEGF, has also been evaluated in advanced HCC patients. A phase II study in advanced HCC with extrahepatic disease found a 14% overall RR with bevacizumab monotherapy.¹⁹³ However, potential adverse events such as arterial/venous thromboembolic events and variceal hemorrhage initially kept bevacizumab from moving to later-stage trials.

In contrast to these negative studies, the multitargeted kinase inhibitor lenvatinib has shown promise as a first-line treatment. Lenvatinib targets VEGFR1-3, FGFR1-4, PDGFR, RET, and c-KIT.¹⁹⁴ Lenvatinib was compared to sorafenib in a first-line treatment setting in a randomized noninferiority phase III trial (REFLECT). In this study, 954 CTP class A patients with a BCLC stage of B or C and an Eastern Cooperative Oncology Group performance status score of 0 or 1 were randomized 1:1 to receive either lenvatinib or sorafenib. OS was similar to sorafenib therapy (13.6 vs 12.3 months; HR = 0.92; 95% CI, 0.79 to 1.06). However, there was a significant improvement in secondary end points including PFS (7.4 vs 3.7 months; HR = 0.66; 95% CI, 0.57 to 0.77), TTP (8.9 vs 3.7 months; HR = 0.63; 95% CI, 0.53 to 0.73), and objective RR (24.1% vs 9.2%; odds ratio = 3.13; 95% CI, 2.15 to 4.56) according to modified RECIST criteria.¹⁹⁵

First-Line Combination Therapies

The addition of cytotoxic chemotherapy or targeted therapy to bevacizumab may augment antitumor activity. Response proportions with various cytotoxic combinations range from 9% to 20%, with disease control rates reportedly as high as 78%.¹⁹⁶ Bevacizumab plus erlotinib may offer enhanced antitumor activity with an RR of 24% and favorable patient outcomes with a median OS of 13.7 months.¹⁹⁷ However, a multicenter, randomized phase II trial of bevacizumab plus erlotinib versus sorafenib monotherapy found no difference in median OS between the two treatments (8.55 vs 8.55 months) (NCT00881751).

The SEARCH trial also confirmed that a combination of erlotinib plus sorafenib provided no additional benefit in HCC. In this

randomized, placebo-controlled, double-blind phase III study, the combination of sorafenib plus erlotinib was compared to sorafenib alone in the first-line setting in 720 advanced HCC patients. There was no statistically significant difference between study arms with regard to the primary end point of OS (combination 9.5 months, sorafenib 8.5 months; HR = 0.93; 95% CI, 0.78 to 1.11).¹⁹⁸

Investigators have also proposed combination strategies with cytotoxic chemotherapy and novel biologic agents to improve the modest results observed with sorafenib. Inhibiting the mitogen-activated protein kinase (MAPK) pathway by sorafenib may restore chemosensitivity by enhancing proapoptotic pathways and dampening multidrug resistance (MDR) pathways. Anthracycline-induced cytotoxicity is mediated by the proapoptotic kinase ASK1.¹⁹⁹ Growth factor-induced MAPK activation via FGF abrogates ASK1 activity. Blockade of the RAF kinases by sorafenib might, therefore, augment the antitumor activity of doxorubicin. Furthermore, MAPK activation leads to the induction of the MDR-1 pump.²⁰⁰ Sorafenib decreases ATP-binding cassette/MDR protein gene expression, restoring HCC sensitivity to doxorubicin in vitro.²⁰¹ Before the approval of sorafenib, doxorubicin was evaluated as monotherapy or in combination with sorafenib in a randomized, double-blind, phase II study involving 96 CTP class A patients.²⁰² PFS increased by ~4 months, and the median OS doubled in favor of combination therapy (13.7 vs 6.5 months; $P = 0.006$). However, cardiac toxicity was notable, with a higher proportion of patients on the combination experiencing left ventricular systolic dysfunction (19% vs 2%). The median cumulative doxorubicin dose was limited to 165 mg/m². The dramatic increase in survival over placebo was striking; however, the lack of sorafenib as a comparator arm limits the interpretation of the trial. Unfortunately, a randomized phase III study of sorafenib versus sorafenib plus doxorubicin in the first-line setting showed a median OS of 9.3 months for sorafenib plus doxorubicin and 9.4 months for sorafenib (HR = 1.05; 95% CI, 0.83 to 1.31).^{203,204} Furthermore, a phase II study of doxorubicin plus sorafenib in the second-line setting after sorafenib failure found no improvement in survival. Therefore, this combination is not recommended in treating advanced HCC patients.²⁰⁵

Similarly, combination chemotherapy such as gemcitabine plus oxaliplatin (GEMOX) had previously demonstrated potential efficacy in HCC,²⁰⁶ and addition of sorafenib to GEMOX might offer synergistic antitumor effects. GEMOX plus sorafenib versus sorafenib was tested in a randomized phase II study (GONEXT).²⁰⁷ The trial enrolled 94 advanced HCC patients. The primary end point was a 4-month PFS of $\geq 50\%$. Unfortunately, the combination of GEMOX plus sorafenib resulted in a 4-month PFS rate of only 61% compared to 54% in the sorafenib monotherapy group. Therefore, the authors did not recommend GEMOX plus sorafenib as a potential standard of care therapy moving forward.

Second-Line Therapies

The interest in developing novel agents for advanced HCC has led to rapidly evolving second-line therapies, with several phase III clinical trials already reported. Regorafenib, a multitargeted kinase inhibitor, was evaluated in a randomized phase III trial versus placebo after progression on sorafenib.²⁰⁸ The study demonstrated a significant improvement in OS of 10.6 months for regorafenib versus 7.8 months for placebo ($P < 0.001$). Regorafenib is approved as second-line therapy for patients who have progressed on sorafenib.

Overexpression of c-MET and its ligand, hepatocyte growth factor, occurs in up to 80% of human HCC tumors.²⁰⁹ Blocking MET with multitargeted kinase inhibitors induces in vitro HCC growth suppression, cell-cycle arrest, and decreased viability, as well as in vivo growth suppression and survival prolongation.²¹⁰ Based on these data, several MET inhibitors are being studied in the setting of advanced HCC.

Tivantinib, a selective MET receptor tyrosine kinase inhibitor, was evaluated at two doses in a randomized, placebo-controlled phase II trial in advanced HCC patients who had progressed after

first-line therapy.²¹¹ For patients with high MET-expressing tumors (defined as $\geq 50\%$ 3 to 4+ expression), tivantinib therapy resulted in longer median TTP (2.7 vs 1.4 months) (HR = 0.43; 95% CI, 0.19 to 0.97) and a longer median OS (7.2 vs 3.8 months) (HR = 0.38; 95% CI, 0.18 to 0.81). These benefits were not observed in patients with low MET-expressing tumors. However, these favorable findings could not be verified in a double-blind, randomized phase III study in advanced HCC patients and high MET-expressing tumors in the second-line setting. Median OS (95% CI) was 8.4 months (6.8 to 10.0) for patients who received tivantinib and 9.1 months (7.3 to 10.4) for patients who received placebo (HR = 0.97; 95% CI, 0.75 to 1.25; $P = 0.81$).²¹²

Cabozantinib, an inhibitor of MET and VEGFR-2, has shown promising efficacy data in a cohort of 41 advanced HCC patients,²¹³ with a 5% confirmed partial RR. Median PFS for the cohort was estimated at 4.2 months and median OS at 15.1 months. The CELESTIAL phase III study compared cabozantinib versus placebo in advanced HCC patients who received sorafenib and progressed on at least one prior therapy (NCT01908426). This was an RCT of 707 patients randomized 2:1 to receive cabozantinib versus placebo. At the second interim analysis, the study was stopped due to a significantly improved OS in patients treated with cabozantinib (median OS of 10.2 months with cabozantinib vs 8.0 months with placebo) (HR = 0.76; $P = 0.005$) as well as PFS (5.2 vs 1.9 months; HR = 0.44; 95% CI, 0.36 to 0.52; $P < 0.001$).²¹⁴ A subgroup analysis of CELESTIAL found that OS and PFS with cabozantinib were highest in patients that had only received first-line sorafenib.²¹⁵

The mammalian target of rapamycin (mTOR) pathway plays a critical role in hepatocarcinogenesis. Blockade of this pathway in xenograft mouse models results in HCC growth suppression and increases survival.²¹⁶ These observations, as well as retrospective data indicating enhanced survival among patients receiving sirolimus immunosuppression following liver transplantation for HCC, led to the development of a phase I/II study of everolimus in patients with advanced HCC, which established that 10 mg/day was a safe dose.²¹⁷ Everolimus was investigated in the second-line setting after sorafenib failure in the phase III, randomized, placebo-controlled EVOLVE-1 study that was reported as negative (NCT01035229).²¹⁸ This was a study of 546 adult patients with BCLC stage B or C HCC and CTP class A liver function who progressed during or following sorafenib or were intolerant to sorafenib. Median survival was 7.6 months with everolimus and 7.3 months with placebo (HR = 1.05; 95% CI, 0.86 to 1.27; $P = 0.68$).

A randomized phase III study of brivanib after progression on sorafenib versus best supportive care also failed to meet its primary end point of improved OS.²¹⁹ Among 395 randomized patients, median OS was 9.4 months for brivanib versus 8.2 months for placebo (HR = 0.89; 95% CI, 0.69 to 1.15; $P = 0.3307$).

Ramucirumab, a monoclonal antibody blocking VEGFR-2, was assessed in a phase II study of 43 patients with systemic treatment-naïve advanced HCC.²²⁰ Median PFS was 4 months and median OS was 12 months. Based on these data, a randomized phase III study (REACH) of ramucirumab versus best supportive care in the second-line setting was performed.²²¹ In this phase III study, 565 patients who were refractory or intolerant to sorafenib were randomly assigned to either ramucirumab or placebo. Unfortunately, second-line treatment with ramucirumab did not significantly improve OS over placebo (HR = 0.87; $P = 0.14$). However, on subgroup analysis, investigators noted a potential survival benefit of ramucirumab in patients with elevated AFP levels ≥ 400 mg/mL. Therefore, the REACH-2 phase III study specifically investigated the role of ramucirumab as a second-line treatment versus placebo, specifically in patients with AFP levels ≥ 400 ng/mL.^{222,223} In this study, 292 patients who had progressed on or were intolerant to sorafenib were randomly assigned 2:1 to receive ramucirumab or placebo. Median OS was increased for ramucirumab compared to placebo (8.5 vs 7.3 months; HR = 0.710; 95% CI, 0.531 to 0.949). This study was the first to demonstrate success in

using a biomarker to identify an HCC patient population more likely to benefit from systemic therapy.

Linifanib (ABD-869) is a receptor tyrosine kinase inhibitor that targets angiogenesis by inhibiting VEGF and PDGF. Linifanib was compared to sorafenib in advanced HCC patients in a negative phase III study.²²⁴ In this randomized study of 1,035 Asian HCC patients, median OS was 9.1 months with linifanib versus 9.8 months with sorafenib (HR = 1.046; 95% CI, 0.896 to 1.22).

Messenger RNA encoding argininosuccinate synthetase is not present in subsets of HCC; therefore, arginine must be extracted from the circulation.²²⁵ Pegylated arginine deiminase (ADI-PEG 20) is an arginine-degrading enzyme isolated from *Mycoplasma* formulated with polyethylene glycol (molecular weight: 20 kDa). A phase I/II study demonstrated an excellent safety profile of ADI-PEG 20 in a patient population with a high disease burden and impaired hepatic function.²²⁶ Of 19 evaluable patients, 2 (10.5%) had a complete response, 7 (36.8%) had a partial response, and 7 (36.8%) had stable disease. The duration of response ranged from 37 to >680 days. However, a double-blind, placebo-controlled phase III study of ADI-PEG 20 after prior systemic therapy was disappointing (median OS of 7.8 months for ADI-PEG 20 vs 7.4 months for placebo) (HR = 1.022; $P = 0.884$).²²⁷

Immunotherapy

Several studies are evaluating the role of checkpoint blockade immunotherapy in HCC, both as first-line and second-line treatments. Nivolumab, an anti-PD-1 antibody, was tested in an HCC-specific open-label phase I/II trial. Patients had progressed on at least one line of systemic therapy, including sorafenib, or were intolerant to sorafenib. Among 262 patients, the safety profile of nivolumab was manageable, and there were two complete responses with an objective RR of 20% (95% CI, 15 to 26),²²⁸ which led to initial accelerated FDA approval of nivolumab in the second-line setting. However, when nivolumab was recently compared to sorafenib as a first-line treatment in a randomized phase III trial (CheckMate 459), there was no significant difference between nivolumab and sorafenib in median OS (16.4 vs 14.7 months; HR = 0.85; 95% CI, 0.72 to 1.02) or PFS (3.7 vs 3.8 months), and statistical significance for the primary endpoint of OS was not achieved.²²⁹ Subsequent long-term follow-up analysis of 33.6 months demonstrated that the OS advantage of nivolumab was increased compared to sorafenib (16.4 vs 14.8 months; HR = 0.85; 95% CI, 0.72 to 1.00), although many patients on the sorafenib arm later received treatment with PD-1 or PD-L1 inhibitors. As a result, the indication for nivolumab in the second-line setting has been voluntarily withdrawn and is no longer recommended by the FDA.²³⁰

The combination of nivolumab plus the CTLA-4 antibody ipilimumab has also been studied in the randomized phase I/II trial within CheckMate 040 as a second-line treatment for advanced HCC patients that previously received sorafenib.²³¹ In this portion of CheckMate 040, patients were randomized 1:1:1 to receive varying combinations of nivolumab plus ipilimumab. The objective RRs in all three cohorts were close to 30%, and the highest median OS was 22.8 months. These results have led to nivolumab plus ipilimumab receiving accelerated approval from the FDA as a second-line treatment for HCC patients. The phase III trial Checkmate 9DW is currently studying nivolumab plus ipilimumab versus SOC (sorafenib or lenvatinib) as a first-line treatment for advanced HCC (NCT04039607).

Similar to nivolumab, pembrolizumab was initially evaluated in an open-label phase II clinical trial (KEYNOTE-224).²³² In this study, 104 advanced HCC patients who had progressed on or were intolerant to sorafenib demonstrated an objective RR of 17% (95% CI, 11 to 26), with a median OS of 12.9 months (95% CI, 9.7 to 15.5). These results led to accelerated FDA approval of pembrolizumab as second-line therapy. Subsequently, investigators reported on a phase III study of pembrolizumab versus best supportive

care in an unselected second-line HCC patient population. In this phase III study, 413 patients were randomly assigned in a 1:1 fashion to pembrolizumab or best supportive care. Although there appeared to be an improvement in survival with pembrolizumab (13.8 vs 10.6 months), it did not meet statistical significance for the prespecified efficacy end point.²³³ The phase III trial KEY-NOTE-394 investigated the efficacy of pembrolizumab compared to placebo in the second-line setting in Asian advanced HCC patients who had previously received sorafenib.²³⁴ An initial report stated that OS was significantly increased in patients that received pembrolizumab compared to placebo; PFS and objective RR were also increased compared to placebo. The use of pembrolizumab as a single-agent second-line therapy remains under question.

In addition to single-agent PD-1/PD-L1 immune checkpoint inhibitors (ICIs), the role of single-agent anti-CTLA4 antibodies has also been explored. In a study evaluating the anti-CTLA4 antibody tremelimumab in advanced HCC patients with HCV-related cirrhosis who had progressed on prior sorafenib, 3 of 17 patients (17.6%) achieved a partial response, and 10 patients (58.8%) had stable disease. Median TTP was 6.48 months (95% CI, 3.95 to 9.14), and median OS was 8.2 months (95% CI, 4.64 to 21.34).²³⁵ Based on phase I data, MEDI4736 (durvalumab), an anti-PD-L1 antibody, and tremelimumab were evaluated as monotherapies and in combination in HCC patients.²³⁶ Patients who had progressed on or were intolerant to sorafenib and had not received previous ICI therapy were recruited. Patients that received durvalumab plus high-dose tremelimumab (300 mg compared to 75 mg) reported a median OS of 18.73 months (95% CI, 10.78 to 27.27) and an objective RR of 24.0% (95% CI, 14.9 to 35.3). As a result, the phase III HIMALAYA study was designed to compare durvalumab plus tremelimumab versus durvalumab versus sorafenib as a first-line treatment in unresectable HCC patients (NCT03298451). Eligible patients must have BCLC stage B or C disease, CTP class A status, and not have received prior systemic therapies. The primary outcome is OS; secondary outcomes include PFS and objective RR. A recent press release stated that high-dose tremelimumab plus durvalumab resulted in a significant increase in OS compared to sorafenib, with a favorable safety profile.²³⁷ There was no increase in severe hepatic toxicity in patients that received tremelimumab plus durvalumab compared to durvalumab alone.

In addition to studying the role of ICIs as single agents or in combination with other ICIs, combination treatment targeting VEGF and PD-L1 has been investigated. Blocking tumor angiogenesis and the immunosuppressive environment it creates via VEGF inhibition may create a more favorable setting for immunotherapy.^{238,239} Recently, the results of IMbrave 150, the phase III clinical trial of the PD-L1 inhibitor atezolizumab plus the anti-VEGF antibody bevacizumab compared to sorafenib in the first-line treatment of unresectable HCC were reported. This was a large multicenter international study, where 336 patients received atezolizumab plus bevacizumab, and 165 patients received sorafenib. Patients who received atezolizumab plus bevacizumab had a significant improvement compared to sorafenib in OS (12 vs 6.8 months) and PFS (6.8 vs 4.3 months). The objective RR was 27.3% (95% CI, 22.5 to 32.5) in the atezolizumab plus bevacizumab arm compared to 11.9% (95% CI, 7.4 to 18.0) in the sorafenib arm. The reduced hazard of death with atezolizumab plus bevacizumab versus sorafenib (HR = 0.58; 95% CI, 0.42 to 0.79; $P < 0.001$) has led to the approval of this combination of immunotherapy and antiangiogenic therapy as a new SOC first-line treatment for advanced HCC.²⁴⁰

In addition to targeting VEGF, several multityrosine kinase inhibitors are being explored in combination with immunotherapy. A phase Ib study of pembrolizumab plus lenvatinib as a first-line treatment for unresectable HCC patients demonstrated anti-tumor activity with a tolerable safety profile.²⁴¹ In this study, 100 CTP class A patients with BCLC stage B or C disease showed an objective RR of 46.0% (95% CI, 36.0 to 56.3) according to modified RECIST criteria, and complete response was observed in 11

patients (11.0%). Median OS was 22.0 months (95% CI, 20.4 to (not reached)). As a result of these promising data, the LEAP-002 phase III study of pembrolizumab plus lenvatinib versus lenvatinib as a first-line treatment for patients with unresectable HCC is ongoing (NCT03713593).²⁴²

In addition to lenvatinib plus pembrolizumab, cabozantinib plus atezolizumab versus sorafenib is being investigated as a first-line treatment for advanced HCC in the phase III trial COSMIC-312 (NCT03755791). Results from the prespecified primary interim analysis demonstrated an advantage in PFS with cabozantinib plus atezolizumab compared to sorafenib; however, the co-primary endpoint of OS was not met.²⁴³ COSMIC-312 is unlikely to meet its co-primary end point of OS. Therefore, how practitioners decide which front-line combination immunotherapy approach to use remains open for discussion.

Immunotherapeutic approaches with T cells are also being studied in HCC. One method is to use T cells modified with a chimeric antigen receptor (CAR). AFP-based CAR T cell therapy is being studied in HCC.²⁴⁴ CAR-glypican-3 (GPC3) T cell therapy is also attractive because GPC3 is highly expressed in HCC with limited expression in normal tissues.²⁴⁵ Recently published results from two phase I studies demonstrated a tolerable safety profile of CAR-GPC3 T cells.²⁴⁶ In these studies, 13 HCC patients that had relapsed at least twice were treated with Y035 CAR-GPC3 T cells. No grade 3 or 4 neurotoxicity or CAR T cell-related infusion reactions were reported. Partial responses were observed in two patients, with an OS of 615 and 385 days and PFS of 111 and 99 days, respectively. One patient had sustained stable disease, with an OS of 1,326 days and PFS of 204 days. All three of these patients experienced significant reductions in serum AFP levels after Y035 CAR-GPC3 T cell treatment. A phase I study to identify the maximum tolerated dose of GPC3-CAR T cells (GLYCART cells) is ongoing (GLYCART) (NCT02905188). Another approach uses autologous T cells. The ongoing phase I/II study AYRA-1 is using ET140203 T cells to target AFP-expressing tumor cells in advanced HCC patients who express the human leukocyte antigen (HLA) A2 and have progressed on or were intolerant to at least two prior systemic therapies (NCT04502082). The phase I/II study AYRA3 will use ECT204 T cells to target GPC3-expressing tumor cells in advanced HCC patients who progressed on or were intolerant to at least two prior systemic therapies; it is expected to open for recruitment shortly (NCT04864054).

Systemic Treatments for Patients with Advanced Liver Cirrhosis

The SHARP and Asia-Pacific studies were limited to unresectable or metastatic HCC patients with preserved liver function. Sorafenib's safety in patients who are not CTP class A is of concern because a subgroup analysis of the phase II sorafenib study reported higher rates of hepatic decompensation, represented by worse hyperbilirubinemia, encephalopathy, and ascites in CTP class B compared to CTP class A patients.²⁴⁷ CTP class B patients had a shorter treatment duration as well as shorter survival compared to CTP class A patients (3.2 vs 9.5 months). Similar findings have been reported from the phase IV GIDEON study of sorafenib.²⁴⁸ CTP class B and C patients were treated with sorafenib for a shorter duration, had shorter median survival, and were more likely to develop hepatic and drug-related adverse events. A phase I study of sorafenib pharmacokinetics in patients with hepatic dysfunction helped establish some guidance on the use of sorafenib among patients with advanced CTP scores.²⁴⁹ For a serum bilirubin $\leq 1.5 \times$ upper limit of normal (ULN), sorafenib at the full dose of 400 mg twice daily can be given, whereas for a bilirubin 1.6 to $3 \times$ ULN, half that dose should be considered. For albumin < 2.5 mg/dL, no more than 200 mg once daily should be given. No safe dose was identified for patients with a bilirubin $> 3 \times$ ULN.

Another multityrosine kinase inhibitor studied in CTP class B patients is cabozantinib. Although the phase III CELESTIAL trial

only enrolled patients with CTP class A status, it tracked outcomes of patients who had progressed to CTP class B within 8 weeks of starting therapy.²⁵⁰ Fifty-one out of 470 patients in the cabozantinib arm and 22 out of 237 patients in the placebo arm had progressed to CTP class B status within 8 weeks. Median OS in the CTP class B subgroup was 8.5 months in the cabozantinib arm compared to 3.8 months in the placebo arm (HR = 0.32; 95% CI, 0.18 to 0.58). Median PFS was greater in the cabozantinib arm compared to the placebo arm (3.7 vs 1.9 months; HR = 0.44; 95% CI, 0.25 to 0.76). There were no significant differences in incidence or type of adverse events between the CTP class B cabozantinib subgroup and the overall cabozantinib arm. This analysis suggests that the safety profile of cabozantinib is potentially tolerable in CTP class B patients and that further studies to evaluate its efficacy within this population are warranted.

In addition to these small molecule inhibitors, ICIs are being evaluated in CTP class B patients. The phase I/II study CheckMate 040 evaluated the safety and efficacy of nivolumab as a single-agent therapy in advanced HCC and included one cohort of CTP class B patients.²⁵¹ Median OS in this cohort was 7.6 months (95% CI, 4.4 to 10.5) and was similar between patients with B7 or B8 scores (7.6 vs 7.4 months). Five patients improved from CTP class B to CTP class A status during treatment, and this improvement was maintained for at least 6 months. Disease control rates were similar in CTP class B patients compared to CTP class A patients (55% vs 61%). There were no significant differences in the incidence of adverse events between CTP class B and class A patients. These results suggest that using nivolumab in CTP class B patients is potentially tolerable and warrants further investigation.

TREATMENT OF OTHER PRIMARY LIVER TUMORS

Fibrolamellar Hepatocellular Carcinoma

Fibrolamellar HCC is a rare primary liver cancer that differs significantly from typical HCC. Fibrolamellar HCC occurs in a much younger age group (peak incidence in the 3rd decade), is usually not associated with cirrhosis or viral hepatitis, and affects males and females equally. Additionally, a much higher percentage (>15%) of fibrolamellar HCC patients presents with positive lymph nodes than normal variant HCC.

Whether the diagnosis of fibrolamellar HCC portends a more favorable outcome after surgical treatment remains controversial.²⁵² Resection is the first line of therapy. For patients in whom the tumor is thought to be unresectable, LT provides an effective alternative. Pinna et al.²⁵² described 13 fibrolamellar HCC patients who received transplantation between 1968 and 1995 and reported 1-, 3-, and 5-year patient survival rates of approximately 90%, 75%, and 38%, respectively. Given that most patients transplanted presented

as young patients with advanced disease, this is not an unacceptable survival rate. El-Gazzaz et al.²⁵³ reported similar survival rates. The results are contrary to what was previously reported in a large 95 patient retrospective analysis, in which median survival was limited to 6.7 years; 50% of patients presented with stage IV disease. Factors significantly associated with poor survival were female sex, advanced stage, lymph node metastases, vascular invasion, and unresectable disease.²⁵⁴

Hepatoblastoma

Hepatoblastoma is the most common primary cancer of the liver occurring in childhood. The annual incidence of hepatoblastoma ranges from 0.5 to 1.5 cases per million children, with a peak incidence occurring within the first 2 years of life.²⁵⁵ Hepatoblastoma is highly sensitive to chemotherapy, which often renders unresectable tumors resectable. Surgical resection is considered the first line of therapy. However, for those tumors that cannot be converted to resectable lesions but without distant metastasis, most can be rescued with LT. Survival rates for these children after LT are excellent, with 1-, 3-, and 5-year survivals reported at 92%, 92%, and 83%, respectively.²⁵⁵

An ongoing partially randomized phase II/III study is evaluating the role of cisplatin as single-agent therapy or in combination with other chemotherapy drugs in pediatric or young adult patients with hepatoblastoma after surgery. Primary outcomes include event-free survival; other outcomes include failure-free survival, OS, and incidence of adverse events (NCT03533582).

Epithelioid Hemangioendothelioma

Epithelioid hemangioendothelioma is a very rare tumor of vascular origin that can originate in the liver; it occurs predominantly in females. The tumor is often confused with other more aggressive cancers, particularly cholangiocarcinoma, angiosarcoma, and HCC. The clinical course of epithelioid hemangioendothelioma is quite variable. In a review by Makhoul et al.,²⁵⁶ 137 cases of epithelioid hemangioendothelioma were described, with survival ranging from 4 months to 28 years. Of interest, one patient who received no treatment survived for 27 years without evidence of metastasis. Surgical resection is considered to be the treatment of choice. However, in multifocal disease, a short observation period may be reasonable to help decide between observation, ablation, resection, or transplant as the course of action. The presence of metastatic disease does not seem to influence survival and should not be considered a contraindication to either resection or transplantation. Systemic therapies that have been used in this patient population include chemotherapies, VEGF inhibitors, and ICIs; however, larger studies of this rare disease are needed to confirm their efficacy and determine the patient subset(s) in which they are most effective.²⁵⁷

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